

Hierarchical Bayesian Mechanism-Adequacy Estimation for Molten Salt Radiolysis Networks: A Practical Framework with Validation Against the Pulse-Radiolysis Literature

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Abstract

Bayesian calibration of chemical-kinetic networks for molten-salt radiolysis must handle heterogeneous data: dense pulse-radiolysis transients, scalar Arrhenius pairs from multiple laboratories, and one-sided detection limits from chronic-irradiation experiments. The same elementary reactions appear in chemically distinct host salts, the reaction-network topology itself is uncertain, and inter-laboratory systematic biases are documented but not formally separated from the intrinsic chemistry. No published framework addresses these features together. We present Hierarchical Bayesian Mechanism-Adequacy Estimation (HBMAE), which combines a constrained-topology prior, a hierarchical Arrhenius layer that pools shared reactions across host salts, a com-

posite likelihood across observation modalities, a parametric Kennedy–O’Hagan discrepancy with bounded flexibility, and an explicit facility-effect hierarchy. Six theorems characterize ergodicity of the topology–coefficient sampler, cross-host posterior consistency with an explicit host-heterogeneity floor, finite-sample prior regularization under parametric discrepancy, sandwich-variance asymptotic normality of the composite estimator, a censored Bayes-factor lower bound for one-sided detection benchmarks, and identifiability of the facility hierarchy. On the Iwamatsu–Horne pulse-radiolysis chloride data set and the Phillips 2022 Molten Chloride Fast Reactor detection-limit benchmark, HBMAE recovers published Arrhenius parameters within reported σ for four chromium and zinc reactions and quantifies a previously qualitative Pikaev–Iwamatsu inter-laboratory offset at $b^{(\text{Pikaev})} = -3.35 [-4.53, -2.44]$. A censored Bayes factor of $\ln \text{BF} = 316.9$ excludes the un-buffered chloride kernel at any plausible significance, given the prior in Section 4.6 and the predictive inputs $(\mu_{\gamma_0} - c)/\sigma_C \approx 25$ of Section 8.3. On a synthetic benchmark with known ground truth, HBMAE recovers the Arrhenius parameters within 1% bias and attains nominal 90% credible-interval coverage; reversible-jump MCMC with weakly informative priors exhibits a ~ 35 kJ/mol E_a bias of the same sign and magnitude as the published-data result. Direct comparison against Kennedy–O’Hagan with Gaussian-process discrepancy, B2BDC, and PSIS-LOO is given in Section 9. A companion paper (Tanore-Tamales, 2026) applies the calibrated framework to construct a multi-salt operational radiolysis model for chloride and fluoride molten-salt reactor fuels.

Keywords: molten salt reactor; radiolysis; chemical kinetic network; Bayesian model adequacy; hierarchical inference; uncertainty quantification; LiCl–KCl eutectic; NaCl–UCl₃.

1 Introduction

The radiation chemistry of molten salts has moved from background concern to front-line engineering question over the past decade as the molten salt reactor (MSR) has re-emerged

as a leading advanced-fission concept. Concept studies for the Molten Chloride Fast Reactor (TerraPower, 2021), the Fluoride-Salt-Cooled High-Temperature Reactor (Forsberg, 2020), and chloride-fueled and fluoride-fueled fast spectrum systems share a common requirement: validated predictions of how the salt’s chemical inventory evolves under radiation, on time scales spanning nanoseconds (radiolytic transient species) through years (cumulative redox shifts and gas-phase product accumulation). Below detection limits, those predictions are required for plant-safety analyses; above detection limits, they govern off-gas system design and corrosion management.

The pulse-radiolysis literature that supplies the underlying rate constants is now substantial. The Brookhaven Linac Electron Accelerator Facility (LEAF) program has produced quantitative kinetics for the solvated electron e_s^- and the dichlorine radical anion $\text{Cl}_2^{\bullet-}$ in molten LiCl–KCl eutectic (Iwamatsu et al., 2022), for chromium-mediated electron transfer (Iwamatsu et al., 2026), for iodide perturbation (Conrad et al., 2023), for neodymium speciation (Castro Baldivieso et al., 2026), and for the first actinide rate constants in chloride media (Rotermund et al., 2024). Earlier Soviet-era work (Pikaev et al., 1982; Makarov et al., 1982; Hagiwara et al., 1987) established the basic transient inventory across the alkali halide series, and recent ab initio molecular dynamics simulations (Kristoffersen and Metiu, 2018) provide thermodynamic-feasibility anchors for the redox sub-network. Steady-state fluorine-gas yields from gamma-irradiated LiF, BeF₂, UF₄, ThF₄, and FLiBe powders have been recently quantified (Davis et al., 2022), complementing the early Toth–Felker recombination measurements (Toth and Felker, 1990). In the chloride-salt nuclear-fuel context, Phillips and coworkers (Phillips et al., 2022) reported the first long-duration gamma-irradiation of NaCl–UCl₃ salt with no detectable Cl₂ above approximately 1000 ppm.

The methodological problem of *combining* these heterogeneous data sources into a predictive radiolysis model has lagged the experimental progress, and that gap is the subject of this paper. Existing Bayesian-calibration frameworks were developed for single-laboratory, single-modality data sets, and the remainder of this section enumerates the concrete ways

they fail to match the radiolysis problem.

1. *Time-scale heterogeneity.* Pulse-radiolysis transients evolve over nanoseconds; chronic-irradiation gas accumulation evolves over months. The ratio is fourteen orders of magnitude. A single stiff-ODE forward solver cannot reach both ends without singular-perturbation reduction.
2. *Observation modality heterogeneity.* Different experimental campaigns report different observable types: dense absorbance time-series, scalar rate constants with quoted σ , derived Arrhenius ($\log A, E_a$) pairs, and one-sided censored detection limits. Treating these as exchangeable Gaussian observations is incorrect for censored data and yields imbalanced posterior contraction across the others.
3. *Cross-laboratory systematic bias.* The Pikaev microsecond pulse radiolysis of the 1980s and the Iwamatsu picosecond LEAF data of the 2020s disagree by an order of magnitude on rate constants for ostensibly the same reaction. Iwamatsu and coworkers have argued qualitatively that Pikaev’s time-resolution introduced a systematic bias (Iwamatsu et al., 2022); no published framework formalizes this as an identifiable parameter.
4. *Cross-salt mechanistic transfer.* The elementary reaction $e_s^- + M^{n+} \rightarrow M^{(n-1)+}$ has been measured in NaCl, KCl, LiCl–KCl, and ZnCl₂ hosts. Activation parameters differ across hosts in chemically-interpretable ways (Marcus-type solvent reorganization), but no published framework uses these as coupled observations of a shared intrinsic chemistry.
5. *Network topology constraints.* Mass and charge balance constrain the feasible set of reaction-inclusion vectors γ to a strict subset of $\{0, 1\}^R$; reaction networks that violate these conservation laws are unphysical. Existing spike-and-slab Bayesian network-inference frameworks (Galagali and Marzouk, 2015, 2019) place positive prior mass on the infeasible region and pay a corresponding mixing penalty.

This paper introduces HBMAE, a Bayesian framework whose components are individually well-established in the uncertainty-quantification literature. The contribution is the synthesis specific to molten-salt radiation chemistry, the identifiability and contraction theorems of Section 5, and the validation against four published data sources.

Section 2 fixes notation and reviews the inference problem. Section 3 surveys the four existing methodological approaches against which HBMAE is compared and formalises the failure modes that motivate the new framework. Section 4 presents HBMAE as a sequence of compositional layers, one per gap above; the section culminates in the unified posterior and a complete sampling algorithm. Section 5 states the theorems and gives full proofs; appendices contain technical lemmas and synthetic-benchmark code. Sections 6 and 7 present two worked examples: a single-reaction illustration of $e_s^- + \text{Cr}^{2+} \rightarrow \text{Cr}^+$ from the Iwamatsu 2026 nanosecond-resolved transients (a minimal use of the framework), and the cross-paper resolution of the Pikaev–Iwamatsu Zn^{2+} inconsistency through the facility-effect and hierarchical layers. Section 8 presents four progressively larger validation case studies with full posterior diagnostics, prior-sensitivity analyses, and Godambe-sandwich verification, culminating in an integrated 24-parameter MCMC across three observation modalities. Section 9 benchmarks HBMAE against four state-of-the-art comparator methods on both a synthetic dataset with known ground truth and the published Zn data. We discuss limitations, open problems, and the connection to the companion application paper (Tanore-Tamale, 2026) in Section 10 and conclude in Section 11.

Code and digitized data are available at [*repository URL to be provided at submission*]. All numerical results in tables and figures are reproducible from the supplied scripts.

2 Background

The reader who is comfortable with stiff-ODE chemical kinetics, Bayesian inference, and model adequacy may skip to Section 3. This section builds the necessary background from

first principles, using molten-chloride radiolysis as the running example.

2.1 Molten-chloride radiolysis as a stiff-ODE system

When ionizing radiation deposits energy in a molten chloride salt, the local chemistry follows a characteristic sequence. In the first few picoseconds, energy deposition generates electron-hole pairs that thermalize and solvate to form the solvated electron e_s^- and the chlorine atom $\text{Cl}^\bullet$. These primary radicals then react with each other and with the salt's constituent species, producing secondary transient species such as the dichlorine radical anion $\text{Cl}_2^{\bullet-}$ and longer-lived products such as the trichloride anion Cl_3^- and molecular chlorine Cl_2 .

For the lithium-potassium chloride eutectic at 400 °C the following five elementary reactions form a minimal kinetic network (Iwamatsu et al., 2022; Hagiwara et al., 1987):



Each k_i depends on temperature through the Arrhenius equation,

$$k_i(T) = A_i \exp\left(-\frac{E_{a,i}}{RT}\right), \quad (6)$$

where A_i is the pre-exponential factor, $E_{a,i}$ is the activation energy, $R = 8.314 \text{ J/mol/K}$, and T is the absolute temperature. The pair $(A_i, E_{a,i})$ is the *closure coefficient* for reaction i ; in this paper we will be inferring posterior distributions on these pairs from data.

The mass-action assumption converts the network (1)–(5) into a system of ordinary

differential equations on the species concentrations,

$$\frac{d\mathbf{C}}{dt} = \mathbf{f}(\mathbf{C}; \boldsymbol{\theta}, \boldsymbol{\gamma}) + \mathbf{S}_{\text{rad}}(t), \quad (7)$$

where $\mathbf{C} \in \mathbb{R}^n$ is the vector of species concentrations, $\boldsymbol{\theta}$ collects the closure coefficients $(\log A_i, E_{a,i})$, $\boldsymbol{\gamma} \in \{0, 1\}^R$ is an inclusion indicator selecting which candidate reactions are in the network, and $\mathbf{S}_{\text{rad}}(t)$ is a source term capturing the radiolytic generation of primary radicals. When $\mathbf{S}_{\text{rad}}$ is a brief pulse, the system models a pulse-radiolysis transient; when $\mathbf{S}_{\text{rad}}$ is constant, the system models chronic irradiation.

The right-hand side of equation (7) is *stiff*: the dominant rate constants are on the order of $10^{10} \text{ M}^{-1} \text{ s}^{-1}$ multiplied by concentrations of order $1 \times 10^{-3} \text{ M}$, giving characteristic inverse-time scales of $\sim 10^7 \text{ s}^{-1}$ for radical disappearance, while the slowest reactions (R_5 , gas-phase escape) operate at $\sim 1 \text{ s}^{-1}$. The seven-order-of-magnitude spread in eigenvalues demands a stiff ODE solver such as BDF (Hindmarsh et al., 2005); this is the first of two scale-separation features that will reappear when we discuss the Phillips chronic-irradiation benchmark in Section 8.

2.2 The inference problem in one sentence

Given observations $\mathbf{y}$ of the species trajectories or derived quantities (Section 2.7), and given a candidate kinetic network specified by the inclusion vector $\boldsymbol{\gamma}$, infer the posterior probability distribution over the closure coefficients $\boldsymbol{\theta}$ and the inclusion vector itself.

This sentence is the entire inference problem. The rest of Section 2 unpacks each technical term in it.

2.3 Probabilistic thinking applied to rate constants

The rate constants in equations (1)–(5) are not known exactly. Even after the most careful pulse-radiolysis measurement, what we have is a measured value with a quoted uncertainty.

For example, Iwamatsu and coworkers report the chromium electron-capture rate as $k_5 = (4.1 \pm 0.2) \times 10^{10} \text{ M}^{-1} \text{ s}^{-1}$ at 400°C (Iwamatsu et al., 2026); the ± 0.2 is one standard deviation of the measurement, and the true value lies in some neighborhood of 4.1×10^{10} that the data alone cannot pin down further.

The Bayesian view of inference takes this neighborhood seriously. Rather than treating k_5 as an unknown number to be pinned to a single value, we treat it as an unknown number described by a *probability distribution*, written $p(k_5)$, whose values quantify our degree of belief in each possible value of k_5 . If the distribution is sharply peaked at 4.1×10^{10} with a narrow width, our uncertainty is small; if the distribution is broad and flat, our uncertainty is large.

The same idea applies to the Arrhenius pair $(\log A_i, E_{a,i})$ for any reaction in the network. We write $p(\boldsymbol{\theta})$ for the joint probability density over all closure coefficients; the joint density encodes both the marginal uncertainty in each individual parameter and the correlation structure between them.

Operations on probability distributions follow the standard rules from a first-course in probability. Most importantly for what follows, two random quantities A and B are characterized by their *joint* distribution $p(A, B)$, and we can recover either marginal by integration,

$$p(A) = \int p(A, B) \text{d}B, \quad p(B) = \int p(A, B) \text{d}A, \quad (8)$$

or write the joint as a product of conditional and marginal:

$$p(A, B) = p(A | B) p(B) = p(B | A) p(A). \quad (9)$$

Equation (9) is the source of Bayes' rule, which is the single technical fact on which everything in the rest of the paper depends.

2.4 Bayes' rule and the three pieces of a posterior

Let $\boldsymbol{\theta}$ denote the closure coefficients we want to infer and let $\mathbf{y}$ denote the data we observed.

Apply equation (9) to the joint $p(\boldsymbol{\theta}, \mathbf{y})$ in two ways and equate them:

$$p(\boldsymbol{\theta} \mid \mathbf{y}) p(\mathbf{y}) = p(\mathbf{y} \mid \boldsymbol{\theta}) p(\boldsymbol{\theta}). \quad (10)$$

Solving for the left-hand factor that we want gives Bayes' rule:

$$\boxed{p(\boldsymbol{\theta} \mid \mathbf{y}) = \frac{p(\mathbf{y} \mid \boldsymbol{\theta}) p(\boldsymbol{\theta})}{p(\mathbf{y})}} \quad (11)$$

The four pieces of equation (11) have names that will recur throughout the paper:

- $p(\boldsymbol{\theta})$ is the *prior*. It encodes what we believed about $\boldsymbol{\theta}$ before seeing $\mathbf{y}$, including any relevant prior literature, physical bounds, and conservation laws.
- $p(\mathbf{y} \mid \boldsymbol{\theta})$ is the *likelihood*. It is the probability density of observing the actual data $\mathbf{y}$ *under the assumption* that the closure coefficients took the value $\boldsymbol{\theta}$. For an ODE model with additive Gaussian observation noise, the likelihood is the product over data points of a Gaussian density centered on the model prediction.
- $p(\boldsymbol{\theta} \mid \mathbf{y})$ is the *posterior* — the new probability distribution over $\boldsymbol{\theta}$ that accounts for the prior and the likelihood together. It is the object we report.
- $p(\mathbf{y}) = \int p(\mathbf{y} \mid \boldsymbol{\theta}) p(\boldsymbol{\theta}) d\boldsymbol{\theta}$ is the *marginal likelihood* or evidence. It is a normalization constant; in most calibration problems it is not directly computed because the posterior is determined by its proportionality to the numerator.

Reading equation (11) aloud: *the posterior is proportional to the prior times the likelihood*.

Where the prior was uncertain and the data is informative, the posterior tightens. Where the prior was already sharp and the data is noisy, the posterior changes little. Where the prior and the data disagree strongly, the posterior compromises in a quantifiable way.

A worked example. Suppose we have measured the chromium electron- capture rate at two temperatures, 400 °C and 500 °C, with values $k_{400} = 4.1 \times 10^{10} \text{ M}^{-1} \text{ s}^{-1}$ and $k_{500} = 9.3 \times 10^{10} \text{ M}^{-1} \text{ s}^{-1}$, each with multiplicative one-sigma noise $\sigma_{\log k} = 0.05$. Suppose the Arrhenius parameters $(\log A, E_a)$ have prior mean $(\log(1.7 \times 10^{13}), 33.5 \text{ kJ/mol})$ and prior covariance $\text{diag}(0.12^2, 600^2)$ (in matching units). Substituting into equation (11), the log-posterior is the sum of two log-Gaussian likelihood terms and two log-Gaussian prior terms,

$$\begin{aligned} \log p(\log A, E_a \mid k_{400}, k_{500}) = & -\frac{1}{2} (\log k_{400} - \log A + E_a/RT_1)^2 / \sigma_{\log k}^2 \\ & -\frac{1}{2} (\log k_{500} - \log A + E_a/RT_2)^2 / \sigma_{\log k}^2 \\ & -\frac{1}{2} (\log A - \log A_{\text{prior}})^2 / \sigma_A^2 \\ & -\frac{1}{2} (E_a - E_{a,\text{prior}})^2 / \sigma_{E_a}^2 + \text{const.} \end{aligned} \quad (12)$$

For Gaussian prior and Gaussian likelihood with a linear model the posterior is itself Gaussian and the mean and covariance can be written down in closed form. Equation (12) is the simplest non-trivial posterior that appears in this paper; HBMAE generalizes the structure to many more parameters, non-Gaussian likelihoods, and multi-level priors.

A natural temptation is to set $\sigma_A = \sigma_{E_a} = \infty$ in the prior on the grounds that this is “maximally uninformative”. This is mathematically improper (the integral of $p(\boldsymbol{\theta})$ diverges) and operationally dangerous: it removes the prior’s ability to break ambiguities in the likelihood, and a likelihood with even mild mis-specification then pulls the posterior to non-physical values. We demonstrate the failure concretely in Section 9 (method M_4 , the Galagali–Marzouk-style weakly-informative analog of HBMAE).

2.5 Posterior computation by Markov chain Monte Carlo

The right-hand side of equation (11) is, in general, an intractable function of $\boldsymbol{\theta}$ because the normalizing constant $p(\mathbf{y})$ is a high-dimensional integral. Markov chain Monte Carlo (MCMC) methods sidestep the normalizing constant by constructing a random walk in $\boldsymbol{\theta}$ -

space whose stationary distribution *is* the unnormalized posterior (Tierney, 1994; van der Vaart, 1998). After discarding an initial transient (“burn-in”), the trajectory of the walk provides samples from the posterior.

The algorithm we use throughout this paper is the affine-invariant ensemble sampler of Goodman and Weare (Goodman and Weare, 2010a), as implemented in `emcee` (Foreman-Mackey et al., 2013). An ensemble of N_{walkers} walkers is initialized in parameter space, and at each step a fraction of the walkers propose new positions based on stretches from the positions of the others. Proposals are accepted with the Metropolis–Hastings probability, which depends only on the ratio of posterior densities at the proposed and current positions. After many iterations, the empirical distribution of walker positions approaches the posterior.

We diagnose convergence by three standard statistics computed across the ensemble (Vehtari et al., 2017, 2021):

- the integrated autocorrelation time τ , which quantifies how many steps must pass before two walker positions are independent;
- the rank-normalised split- $\hat{R}$ of Vehtari et al. (2021), which compares within-walker and between-walker variance of any scalar function of $\boldsymbol{\theta}$ and should approach unity as the chain mixes; the rank-normalised variant is robust to heavy-tailed marginals where the classical Gelman–Rubin $\hat{R}$ can fail silently.
- the effective sample size, defined as the total number of samples divided by τ , which should be $\gtrsim 500$ per parameter for stable tail credible-interval estimation (Vehtari et al., 2021).

When all three diagnostics pass, we summarize the posterior with the posterior median and the central 95% credible interval, computed respectively as the 50th and the 2.5th/97.5th percentiles of the samples for each parameter.

2.6 Model adequacy and the discrepancy term

Bayes’ rule presumes that the data $\mathbf{y}$ were generated by the model $p(\mathbf{y} \mid \boldsymbol{\theta})$ for some value of $\boldsymbol{\theta}$. For computer models of physical systems, this presumption is almost always false: the model omits chemistry, simplifies geometry, or uses approximate constitutive relationships. Kennedy and O’Hagan formalized this *model discrepancy* (Kennedy and O’Hagan, 2001) by writing

$$\mathbf{y}_i = \zeta(x_i; \boldsymbol{\theta}^*) + \delta(x_i) + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \sigma^2), \quad (13)$$

where ζ is the computer model output, $\boldsymbol{\theta}^*$ is the “true” value of the calibration parameters, δ is a discrepancy function capturing model inadequacy, and ε is observation noise.

Equation (13) has a known pathology, identified by Brynjarsdóttir and O’Hagan (Brynjarsdóttir and O’Hagan, 2014): if δ is modeled as a nonparametric Gaussian process with stationary covariance, then the posterior on $\boldsymbol{\theta}^*$ is biased with the bias not vanishing as $n \rightarrow \infty$. Their three proposed mitigations are (i) physics-informed prior constraints on δ , (ii) staged calibration with δ estimated on a subset of data where the model is known to be adequate (Vollmer, 2009; Plumlee, 2017), and (iii) replacing the nonparametric δ with a constrained parametric form. An alternative asymptotic remedy is the L^2 -orthogonalised calibration of Tuo and Wu (2015, 2016), which removes the score-aligned-discrepancy bias by projection but is computationally heavier on stiff-ODE forward models. HBMAE adopts the parametric strategy: we use a low-dimensional parametric δ , bounded by physics, and we prove (Section 5, Theorem 4) that the residual bias is controllable in terms of the prior strength.

2.7 Heterogeneous radiation-chemistry observations

A central feature of the radiation-chemistry calibration problem is that the data come in four qualitatively different modalities, each requiring a different likelihood factor:

(M1) Transient absorbance time-series. Pulse-radiolysis experiments deliver a brief electron pulse and record absorbance at a selected wavelength as a function of time over the ensuing nanoseconds to microseconds. Each trace is a vector of (t_j, A_j) pairs, related to the model via Beer’s law $A_j = \varepsilon_\lambda \ell C_q(t_j; \boldsymbol{\theta})$, where ε_λ is the molar absorptivity of the absorbing species q at the probe wavelength λ , ℓ is the optical path length, and C_q is the predicted concentration. The likelihood factor is a product of per-point Gaussians.

(M2) Scalar rate constants. Many experiments report a single rate constant $k_i(T)$ at a single temperature, often as the slope of a pseudo-first-order plot. The likelihood factor is a single log-Gaussian on $\log k_i$ with the reported uncertainty.

(M3) Censored detection bounds. Some experiments report only an upper or lower bound on a species concentration, typically the detection limit of the analytical instrument used. The relevant likelihood factor is a normal cumulative distribution function evaluated at the bound, formally the same as Tobin’s (Tobin, 1958) censored-regression likelihood.

(M4) Derived Arrhenius pairs. Some experiments report the $(\log A_i, E_{a,i})$ pair derived from a multi-temperature fit, with a joint covariance. The likelihood factor is a bivariate Gaussian.

The Godambe-balanced composite likelihood (Section 4, equation (17)) combines the four modalities into a single posterior, weighted to prevent any one modality from dominating purely by virtue of having more observations.

3 State-of-the-art comparator methods

We benchmark HBMAE against four published frameworks for Bayesian calibration of chemical reaction networks. Each is a defensible choice for some single-modality, single-laboratory calibration problem, and each leaves at least one of the five gaps of Section 1 unaddressed.

We summarize each here and return to a quantitative head-to-head comparison in Section 9.

3.1 Galagali–Marzouk reversible-jump MCMC

Galagali and Marzouk (Galagali and Marzouk, 2015) introduced a Bayesian framework for joint inference over the reaction-inclusion vector $\gamma \in \{0, 1\}^R$ and the rate constants θ . A point-mass mixture prior places probability π_i on $\gamma_i = 1$ (reaction i active, with $\log A_i$ drawn from a literature-informed Gaussian) and probability $1 - \pi_i$ on $\gamma_i = 0$ (reaction inactive). Inference uses reversible-jump Markov chain Monte Carlo (Green, 1995) — or, in alternative formulations, the transitional MCMC of Ching and Chen (2007) — to sample jointly the network topology and the parameters of the active reactions, with proposals adapted online via expectation-maximization. A follow-up (Galagali and Marzouk, 2019) exploits network topology to accelerate the between-model moves on networks of size $R \sim 50$, and the local-approximation accelerator of Conrad et al. (2016) can be combined with either sampler to reduce the per-step ODE-solve cost.

The Galagali–Marzouk framework is the closest published precedent to ours, and our constrained-topology prior in Section 4.2 extends their construction with mass and charge balance constraints. We note that the Galagali–Marzouk 2019 extension does in principle support topology constraints via a custom prior over γ ; the differentiator of HBMAE is the *integration* of mass/charge balance as the first layer of inference rather than as a post-hoc prior trim. The two gaps the framework leaves open are: the absence of a hierarchical layer to couple shared reactions across hosts (gap 4 of Section 1), and the absence of facility-effect terms for cross-laboratory bias (gap 3).

3.2 Kennedy–O’Hagan calibration with Gaussian-process discrepancy

The Kennedy–O’Hagan framework (Kennedy and O’Hagan, 2001) treats the model output and the discrepancy as two separate Gaussian processes (one over $\boldsymbol{\theta}$, one over x), and infers both jointly from the data. The framework is widely applied across the physical sciences and is the canonical reference for “model calibration with discrepancy”.

Two limitations are relevant for our problem. First, the discrepancy GP introduces identifiability pathology under mild conditions on the covariance structure, as we discussed in Section 2.6. HBMAE uses Brynjarsdóttir and O’Hagan’s parametric mitigation (Brynjarsdóttir and O’Hagan, 2014). Second, the framework assumes a single observation modality and does not naturally combine the four modalities of Section 2.7.

3.3 B2BDC deterministic uncertainty quantification

The Bound-to-Bound Data Collaboration framework of Frenklach and coworkers (Frenklach, 2007; Hegde et al., 2018; Oreluk et al., 2021) is a non-Bayesian alternative. Each experimental observation is treated as a deterministic constraint on the parameter space, of the form $\ell_i \leq f_i(\boldsymbol{\theta}) \leq u_i$. The feasible set is the intersection of all such constraints; any parameter vector in the feasible set is consistent with all data simultaneously. When the feasible set is empty, the dataset is jointly inconsistent and the framework identifies which constraints are responsible.

B2BDC is a useful preprocessing tool: before any Bayesian inference, the B2BDC consistency check identifies inter-paper disagreements that no posterior can paper over. Section 8 reports a B2BDC-style consistency check on the Pikaev–Iwamatsu Zn rate data that returns $\chi^2 = 96.5$ on five degrees of freedom ($p < 10^{-17}$), formalizing the qualitative critique of Iwamatsu et al. (Iwamatsu et al., 2022).

3.4 Vehtari PSIS-LOO for model comparison

Pareto-smoothed importance-sampling leave-one-out cross-validation (PSIS-LOO) (Vehtari et al., 2017) estimates the expected log predictive density on held-out data without re-running MCMC for each held-out point. PSIS-LOO is the canonical Bayesian model-comparison tool today. It does not constitute a calibration framework in itself but is useful for choosing between candidate calibration models; we use it to rank the comparator methods in Section 9.

3.5 Summary of capabilities

Table 1 maps the five capability requirements of Section 1 against the four comparator methods plus HBMAE. HBMAE is the only framework with full coverage; each comparator handles some subset.

Table 1: Capability coverage of HBMAE versus four state-of-the-art comparator methods. ● = the framework includes this capability natively; ○ = the capability can be retrofitted with additional machinery; – = not provided.

Capability	GM 2015/19	KO 2001	B2BDC	PSIS-LOO	HBMAE
Multi-modality composite likelihood	–	–	–	–	●
Cross-host (cross-salt) hierarchy	–	–	–	–	●
Facility-effect bias separation	–	–	–	–	●
Censored / one-sided observations	–	○	●	–	●
Topology constraints (mass, charge)	○	–	●	–	●
Parametric discrepancy	–	○	●	–	●

4 The HBMAE framework

We now present the HBMAE framework as a composition of five layers, one per gap of Section 1. Each layer is mathematically a single contribution to the posterior; the unified posterior is the product of the five contributions and the likelihood.

4.1 Notation

We collect the principal symbols in Table 2. All concentrations are in mol/m³ (SI); rate constants follow the same unit system except where literature values are explicitly noted in M⁻¹ s⁻¹.

Table 2: Principal notation.

Symbol	Meaning
$\mathcal{S}$	set of salt hosts; $K = \mathcal{S} $
$\mathcal{R}$	set of candidate elementary reactions; $R = \mathcal{R} $
$\gamma \in \{0, 1\}^R$	reaction-inclusion vector
$\Gamma \subset \{0, 1\}^R$	feasibility set; satisfies mass and charge balance
$\theta \in \mathbb{R}^{2R}$	closure coefficients ($\log A_i, E_{a,i}$) for each reaction
$\theta_i^{(s)}$	host-specific Arrhenius for reaction i in salt s
$\eta_i^{(s)}$	host-specific perturbation: $\theta_i^{(s)} = \theta_i + \eta_i^{(s)}$
Λ_i	hyperprior covariance on $\eta_i^{(s)}$
T	absolute temperature, K (used throughout the Arrhenius law)
$b^{(p)}$	facility-effect offset for paper p
ω	parameters of the model-discrepancy field δ
$\mathcal{D}$	assembled data set, partitioned by modality $m \in \{T, K, C, A\}$
w_m	Godambe weight for modality m in the composite likelihood
$\pi(\cdot)$	prior densities; $p(\cdot)$ posterior or conditional densities

4.2 Layer 1 — Constrained-topology prior

Mass and charge balance impose linear constraints on the inclusion vector γ . Define the stoichiometric matrix $\mathbf{S}$ on the non-conserved species (excluding the eutectic anion Cl^- in chloride hosts, which is in vast excess and effectively constant) and require $\mathbf{S}\gamma$ to lie in the column-span of the matrix of radiolytic source terms. The feasibility set is

$$\Gamma = \{\gamma \in \{0, 1\}^R : \mathbf{S}\gamma \in \text{span}(\mathbf{S}_{\text{src}}), \text{ cycle condition holds}\}, \quad (14)$$

where the cycle condition requires every reaction in γ to have a substrate-production and product-consumption pathway within γ . The constrained spike-and-slab prior is

$$\pi_{\Gamma}(\gamma) \propto \mathbf{1}[\gamma \in \Gamma] \prod_{i=1}^R \rho_i^{\gamma_i} (1 - \rho_i)^{1-\gamma_i}. \quad (15)$$

For the LiCl-KCl + Cr kernel of this paper ($R = 10$ candidate reactions), exhaustive enumeration yields $|\Gamma| = 392$ out of 1024 networks (38.3% feasibility). We prove in Lemma 1 that Γ is connected under two-flip moves on the inclusion vector, which guarantees ergodicity of the reversible-jump sampler over the constrained space.

4.3 Layer 2 — Hierarchical Arrhenius across salts

For each reaction i that appears in multiple salt hosts, we decompose the host-specific Arrhenius parameters as

$$\boldsymbol{\theta}_i^{(s)} = \boldsymbol{\theta}_i + \boldsymbol{\eta}_i^{(s)}, \quad \boldsymbol{\eta}_i^{(s)} \sim \mathcal{N}(\mathbf{0}, \Lambda_i) \text{ (independent across } s), \quad (16)$$

where $\boldsymbol{\theta}_i$ is the *intrinsic* (host-independent) Arrhenius pair and $\boldsymbol{\eta}_i^{(s)}$ is the host-specific perturbation. This is the standard hierarchical-Bayes / multilevel structure (Gelman et al., 2013, Ch. 5) applied to chemically-similar host salts. The hyperprior covariance Λ_i is set by Marcus-type considerations (Marcus, 1965): solvent reorganization shifts the activation energy by $O(kT)$ per host, suggesting $\Lambda_i \sim \text{diag}(0.5^2, (5 \text{ kJ/mol})^2)$ in $(\log A, E_a)$ coordinates. Theorem 3 shows that the marginal posterior on $\boldsymbol{\theta}_i$ has two regimes: it contracts with total within-host information when measurement error dominates, but reaches a host-heterogeneity floor when chemistry differences dominate.

4.4 Layer 3 — Godambe-balanced composite likelihood

For the four observation modalities $m \in \{T, K, C, A\}$ of Section 2.7, the composite log-likelihood is

$$\ell_{\text{GC}}(\boldsymbol{\theta}) = \sum_{m \in \{T, K, C, A\}} w_m \log L_m(\boldsymbol{\theta}), \quad w_m = \frac{\{\text{tr}(\hat{I}_m) + \epsilon_I\}^{-1}}{\sum_{m'} \{\text{tr}(\hat{I}_{m'}) + \epsilon_I\}^{-1}}, \quad (17)$$

where $\hat{I}_m = -\nabla^2 \log L_m(\hat{\boldsymbol{\theta}}_{\text{MAP}})$ is the *observed* information matrix of modality m evaluated at the MAP estimate (a plug-in for the unknown Fisher information at the truth), and $\epsilon_I > 0$ is a numerical ridge (set to $10^{-6} \times \min_m \text{tr}(\hat{I}_m)$ in the reference implementation) that prevents division by zero for modalities with rank-deficient observed information. These weights are not claimed to be universally variance-optimal. They are a deliberate information-balancing rule: a dense transient trace should not dominate the posterior purely because it contributes many correlated pseudo-observations, and a single censored detection limit should not be numerically erased by the trace length of a different modality. Because $\hat{I}_m$ is itself a random object the weights w_m are also random; under the regularity conditions of Theorem 5 (see Remark 1), $\hat{I}_m \rightarrow I_m$ a.s. and by Slutsky's theorem the asymptotic Godambe sandwich covariance is the same as the deterministic-weight limit. Theorem 5 then supplies the correct Godambe sandwich covariance for whatever positive (possibly random) weights are used.

The modality-specific likelihood factors are:

$$\log L_T = -\frac{1}{2} \sum_j \frac{(\log A_j^{(s,p)} - \log[\varepsilon_\lambda \ell C_q^{(s)}(t_j; \boldsymbol{\theta})] - b_T^{(p)} - \delta_j)^2}{\sigma_T^2}, \quad (18)$$

$$\log L_K = -\frac{1}{2} \sum_i \frac{(\log k_i^{(s,p)} - \log k_i^{(s)}(T) - b_K^{(p)} - \delta_i)^2}{\sigma_K^2}, \quad (19)$$

$$\log L_C = \log \Phi \left(\frac{c - \mu_C(\boldsymbol{\theta}) - b_C^{(p)}}{\sigma_C} \right), \quad (20)$$

$$\log L_A = -\frac{1}{2} (\boldsymbol{\theta}_i^{(s)} - \boldsymbol{\theta}_i^{\text{pub}})^\top V_{\text{pub}}^{-1} (\boldsymbol{\theta}_i^{(s)} - \boldsymbol{\theta}_i^{\text{pub}}). \quad (21)$$

4.5 Layer 4 — Parametric model-discrepancy

We adopt a parametric discrepancy in the design variables temperature and log-concentration:

$$\begin{aligned} \delta(T, \log[M]; \boldsymbol{\omega}) &= \omega_0 + \omega_1 (1/T - 1/T_{\text{ref}}) \\ &\quad + \omega_2 (\log[M] - \log[M]_{\text{ref}}), \end{aligned} \tag{22}$$

with $\boldsymbol{\omega} \sim \mathcal{N}(\mathbf{0}, \text{diag}(\tau_j^2))$ and τ_j chosen such that $|\delta| \leq B$ with prior probability 0.99 at the boundary of the experimental design (typically $B = 0.2$ on log observables, i.e. 20% in concentration units). Three coefficients suffice to capture the dominant unmodeled physics in pulse radiolysis (Arrhenius- shifted impurity scavenging, concentration-dependent self-recombination). Theorem 4 characterizes the asymptotic bias on $\boldsymbol{\theta}$ under this parametric form.

4.6 Layer 5 — Facility-effect bias terms

For each paper p and each modality m , we admit an additive offset $b_m^{(p)}$ on the log-observable scale (this is the modality-indexed form used throughout the joint posterior of equation (24); where the modality index is clear from context we write $b^{(p)}$ for brevity). The prior is

$$b_m^{(p_{\text{ref}})} \equiv 0, \quad b_m^{(p)} \sim \mathcal{N}(0, \Sigma_b) \text{ for } p \neq p_{\text{ref}}, \tag{23}$$

where p_{ref} is a designated reference laboratory (in our application: Iwamatsu LEAF picosecond pulse radiolysis). The gauge-fixing $b^{(p_{\text{ref}})} \equiv 0$ is required for identifiability; see Theorem 7. The hyperprior variance Σ_b is set wide ($\sigma_b = \log(10)/2 \approx 1.15$) to cover the order-of-magnitude disagreement between Pikaev-era microsecond pulse radiolysis and Iwamatsu-era picosecond LEAF.

4.7 The unified posterior

Combining the five layers and the modality-specific likelihoods, the joint posterior is

$$\begin{aligned}
p(\boldsymbol{\gamma}, \boldsymbol{\theta}, \{\eta^{(s)}\}, \{b_m^{(p)}\}, \boldsymbol{\omega} \mid \mathcal{D}) &\propto \pi_{\Gamma}(\boldsymbol{\gamma}) \prod_i \pi(\boldsymbol{\theta}_i) \\
&\times \prod_s \prod_{i: \gamma_i=1} \mathcal{N}(\eta_i^{(s)} \mid \mathbf{0}, \Lambda_i) \prod_{p \neq p_{\text{ref}}} \prod_m \mathcal{N}(b_m^{(p)} \mid 0, \Sigma_b) \quad (24) \\
&\times \pi(\boldsymbol{\omega}) \exp[\ell_{\text{GC}}(\boldsymbol{\theta}, \eta, b, \boldsymbol{\omega}; \mathcal{D})].
\end{aligned}$$

4.8 Algorithm and implementation

Algorithm 1 specifies the constrained-hierarchical-censored reversible-jump Gibbs–MH sampler for the joint posterior (24). Each sweep partitions the state into topology, kinetic parameters, host perturbations, facility offsets, and discrepancy coefficients. The blocks are updated in sequence, and the per-modality log-likelihoods are cached once per state for diagnostic reuse.

Algorithm 1 HBMAE Gibbs–MH sampler

Require: Data $\{\mathcal{D}^{(m)}\}_{m=1}^M$ partitioned by modality; candidate set $\mathcal{R}_{\text{cand}}$ with feasibility set Γ ; prior hyperparameters $\boldsymbol{\eta}$; initial state $(\boldsymbol{\gamma}^{(0)}, \boldsymbol{\theta}^{(0)}, \boldsymbol{\eta}^{(0)}, \mathbf{b}^{(0)})$ with $b^{(p_{\text{ref}})} \equiv 0$.

Ensure: T posterior samples $\{(\boldsymbol{\gamma}^{(t)}, \boldsymbol{\theta}^{(t)}, \boldsymbol{\eta}^{(t)}, \mathbf{b}^{(t)})\}_{t=1}^T$.

- 1: **for** $t = 1, \dots, T$ **do**
 - 2: **TOPOLOGY STEP** (RJ-MH): propose $\boldsymbol{\gamma}' \sim q(\cdot \mid \boldsymbol{\gamma}^{(t-1)})$ supported on two-flip neighbours of $\boldsymbol{\gamma}^{(t-1)}$ in Γ ; accept with probability $\min(1, \alpha_{\boldsymbol{\gamma}})$ per Eq. (25).
 - 3: **CLOSURE STEP** (WITHIN-MODEL EMCEE): for each ensemble walker $w = 1, \dots, W$, propose stretch move on $\boldsymbol{\theta}_w$ holding $\boldsymbol{\gamma}^{(t)}$ fixed; accept with the affine-invariant rule (26).
 - 4: **HYPERPRIOR STEP**: draw $\boldsymbol{\eta}^{(t)}$ from its conjugate full conditional given $\{\boldsymbol{\theta}^{(s)}\}_s$, using (27)–(28).
 - 5: **FACILITY STEP**: draw $\mathbf{b}^{(t)}$ from its conditional given the reference, centering, or calibration convention of Theorem 7, using (29)–(31).
 - 6: **COMPOSITE LIKELIHOOD EVALUATION**: cache the per-modality log-likelihood $\ell_m(\boldsymbol{\theta}^{(t)})$ once per state for diagnostic reuse and for the Godambe weights of Eq. (17).
 - 7: **end for**
 - 8: **return** $\{(\boldsymbol{\gamma}^{(t)}, \boldsymbol{\theta}^{(t)}, \boldsymbol{\eta}^{(t)}, \mathbf{b}^{(t)})\}_{t=1}^T$ and cached $\{\ell_m^{(t)}\}$.
-

Acceptance ratio for the topology step (RJ-MH). Let $\pi(\boldsymbol{\gamma}, \boldsymbol{\theta}_{\boldsymbol{\gamma}})$ denote the joint prior on the model index and its in-model parameters and let $L(\mathcal{D} \mid \boldsymbol{\gamma}, \boldsymbol{\theta}_{\boldsymbol{\gamma}})$ denote the Godambe-weighted composite likelihood. Because the two-flip move preserves the dimension of the active parameter block for the two flipped reactions (one is born from its prior, one is killed against its prior), the Jacobian of the dimension-matching transformation is unity and the standard reversible-jump ratio collapses to

$$\alpha_{\boldsymbol{\gamma}} = \frac{L(\mathcal{D} \mid \boldsymbol{\gamma}', \boldsymbol{\theta}_{\boldsymbol{\gamma}'})}{L(\mathcal{D} \mid \boldsymbol{\gamma}^{(t-1)}, \boldsymbol{\theta}_{\boldsymbol{\gamma}^{(t-1)}})} \cdot \frac{\pi_{\Gamma}(\boldsymbol{\gamma}') \pi(\boldsymbol{\theta}_{\boldsymbol{\gamma}' \setminus \boldsymbol{\gamma}^{(t-1)}})}{\pi_{\Gamma}(\boldsymbol{\gamma}^{(t-1)}) \pi(\boldsymbol{\theta}_{\boldsymbol{\gamma}^{(t-1)} \setminus \boldsymbol{\gamma}'})} \cdot \frac{q(\boldsymbol{\gamma}^{(t-1)} \mid \boldsymbol{\gamma}')}{q(\boldsymbol{\gamma}' \mid \boldsymbol{\gamma}^{(t-1)})}. \quad (25)$$

The proposal kernel q is uniform over the two-flip neighbours of $\boldsymbol{\gamma}^{(t-1)}$ in Γ , which (by Lemma 1) is non-empty for every $\boldsymbol{\gamma} \in \Gamma$; the proposal ratio is the ratio of neighbour-set cardinalities.

Acceptance ratio for the within-model emcee step. For walker w at state $\boldsymbol{\theta}_w^{(t-1)}$ and a randomly drawn “complement” walker $\boldsymbol{\theta}_v^{(t-1)}$ ($v \neq w$), the affine-invariant stretch proposal is $\boldsymbol{\theta}'_w = \boldsymbol{\theta}_v + Z(\boldsymbol{\theta}_w^{(t-1)} - \boldsymbol{\theta}_v)$ with Z drawn from a density $g(z) \propto 1/\sqrt{z}$ on $[1/a, a]$ ($a = 2$ by default). The affine-invariant acceptance probability of Goodman and Weare (2010b) is

$$\alpha_{\text{emcee}} = Z^{d-1} \frac{p(\boldsymbol{\theta}'_w \mid \boldsymbol{\gamma}^{(t)}, \mathcal{D})}{p(\boldsymbol{\theta}_w^{(t-1)} \mid \boldsymbol{\gamma}^{(t)}, \mathcal{D})}, \quad (26)$$

where d is the active-parameter dimension under $\boldsymbol{\gamma}^{(t)}$ and the Z^{d-1} factor accounts for the volume element of the stretch.

Hyperprior full conditional. Under the Gaussian hierarchy $\boldsymbol{\theta}_i^{(s)} \sim \mathcal{N}(\boldsymbol{\mu}_i, \text{diag}(\sigma_i^2))$ with conjugate priors $\boldsymbol{\mu}_i \sim \mathcal{N}(\boldsymbol{\mu}_0, \Sigma_0)$ and $\sigma_i^2 \sim \text{Inv-Gamma}(\alpha_0, \beta_0)$ (independently per compo-

nent), the full conditionals are conjugate:

$$\begin{aligned} \boldsymbol{\mu}_i \mid \cdots &\sim \mathcal{N}\left(\Sigma_{\mu_i}\left(\Sigma_0^{-1}\boldsymbol{\mu}_0 + \text{diag}(\sigma_i^{-2})\sum_{s=1}^K\boldsymbol{\theta}_i^{(s)}\right), \Sigma_{\mu_i}\right), \\ \Sigma_{\mu_i}^{-1} &= \Sigma_0^{-1} + K \text{diag}(\sigma_i^{-2}), \end{aligned} \quad (27)$$

$$\sigma_{i,j}^2 \mid \cdots \sim \text{Inv-Gamma}\left(\alpha_0 + \frac{K}{2}, \beta_0 + \frac{1}{2}\sum_{s=1}^K(\theta_{i,j}^{(s)} - \mu_{i,j})^2\right). \quad (28)$$

When the hierarchy is parameterized by an unrestricted covariance Λ_i rather than a diagonal one, a Hamiltonian Monte Carlo step on the log-Cholesky factor replaces (28).

Facility full conditionals under reference, centering, or calibration anchors. Stacking the modality-aggregated facility data as $\mathbf{y}^{(p)} = (\log k_i^{(p)} - \log k_i^{(s)}(T_i; \boldsymbol{\theta}))_i$ with precision $\tau_p = n_p/\sigma_K^2$ and prior precision $\tau_b = \Sigma_b^{-1}$, the conditional updates are:

$$(a) \text{ reference anchor: } b^{(p_{\text{ref}})} = 0; b^{(p)} \mid \cdots \sim \mathcal{N}\left(\frac{\tau_p \bar{y}^{(p)}}{\tau_p + \tau_b}, \frac{1}{\tau_p + \tau_b}\right), p \neq p_{\text{ref}}, \quad (29)$$

$$(b) \text{ sum-to-zero centering: } \mathbf{b} \mid \cdots \sim \mathcal{N}\left(\boldsymbol{\mu}_b - D_b \mathbf{1} \frac{\mathbf{1}^\top \boldsymbol{\mu}_b}{\mathbf{1}^\top D_b \mathbf{1}}, D_b - D_b \mathbf{1} (\mathbf{1}^\top D_b \mathbf{1})^{-1} \mathbf{1}^\top D_b\right), \quad (30)$$

$$(c) \text{ calibration anchor: } b^{(p)} \mid \cdots \sim \mathcal{N}\left(\frac{\tau_p \bar{y}^{(p)} + \tau_{\text{cal}} y_{\text{cal}}^{(p)}}{\tau_p + \tau_{\text{cal}} + \tau_b}, \frac{1}{\tau_p + \tau_{\text{cal}} + \tau_b}\right), \forall p, \quad (31)$$

where $\mu_{b,p} = \tau_p \bar{y}^{(p)} / (\tau_p + \tau_b)$, $D_b = \text{diag}\{(\tau_p + \tau_b)^{-1}\}_{p=1}^P$, and τ_{cal} is the precision of the calibration factor of Assumption (A6.2)(c). A bounded prior on $\mathbf{b}$ can be useful for regularization after one of these conventions is chosen, but bounded support alone does not identify the absolute log-rate origin.

Per-step complexity.

1. *Topology step:* $O(|\mathcal{N}_2(\gamma)|)$ proposal- set enumeration plus one composite-likelihood evaluation $O(C_{\text{ODE}})$, where C_{ODE} is the dominant cost of the stiff BDF forward solve.

Across $R = 10$ candidate reactions the neighbourhood is bounded by $\binom{R}{2} = 45$ and feasibility filtering in Γ typically retains ≈ 10 –15 moves.

2. *Closure step*: W ensemble walkers $\times$ one ODE solve per walker per sub-step, $O(W C_{\text{ODE}})$; with $W = 56$ this is the dominant per-sweep cost.
3. *Hyperprior step*: conjugate draws are $O(d^2 K)$ for the $\boldsymbol{\mu}_i$ update and $O(K)$ for each $\sigma_{i,j}^2$; sub- dominant to the closure step.
4. *Facility step*: $O(P)$ Gaussian draws under (a) and (c), $O(P)$ truncated-Gaussian draws under (b); negligible cost.
5. *Composite-likelihood evaluation*: one $O(C_{\text{ODE}})$ stiff ODE solve and $O(M)$ modality reductions; cached once per state so that WAIC and PSIS-L00 diagnostics reuse the trace without recomputation.

The reference implementation in this paper uses Python with the affine- invariant ensemble sampler of `emcee` (Foreman-Mackey et al., 2013). The stiff ODE forward solve uses SciPy’s BDF method (Hindmarsh et al., 2005) with absolute tolerance 10^{-15} and relative tolerance 10^{-9} , adaptive to LSODA fallback if BDF fails. For the integrated run of Section 8, 56 walkers run for 200 production steps after 100 burn-in steps; total runtime is ≈ 25 minutes on a single core. Alternative software stacks (PyMC with `sunode`, Stan with `Torsten`, or Julia’s `DifferentialEquations.jl` with `Turing.jl`) are equivalent in expressiveness but differ in ODE-solver maturity and gradient backpropagation; see SciML Open Source Scientific Machine Learning (2024) for a current survey.

5 Theorems and plain-language interpretations

We collect the six theorems that characterize the well-posedness, identifiability, and asymptotic behavior of the HBMAE framework. For each theorem we list the regularity assumptions explicitly, state the formal conclusion, and (where the proof fits) provide a self-contained

argument in the main text. Two of the proofs (Theorem 4 on discrepancy bias and Theorem 5 on Godambe sandwich asymptotics) are long enough to be relegated to Appendix A; the remaining four proofs are given here.

5.1 Theorem 1 — Ergodicity of constrained RJ-MCMC

We work on the product state space $\mathcal{X} = \Gamma \times \mathbb{R}^d$ with $\Gamma \subset \{0, 1\}^R$ the feasibility set of equation (14) and $\mathbb{R}^d$ the within-model continuous parameter space. Let $\pi(\boldsymbol{\gamma}, \boldsymbol{\theta}) \propto p(\mathcal{D} \mid \boldsymbol{\gamma}, \boldsymbol{\theta}) \pi_\Gamma(\boldsymbol{\gamma}) \pi(\boldsymbol{\theta} \mid \boldsymbol{\gamma})$ denote the target posterior, with π_Γ the constrained spike-and-slab of equation (15). Write $\mathcal{N}_2(\boldsymbol{\gamma}) = \{\boldsymbol{\gamma}' \in \{0, 1\}^R : \|\boldsymbol{\gamma}' - \boldsymbol{\gamma}\|_1 \leq 2\}$ for the two-flip neighborhood.

Assumption 1. **(A1.1)** The data likelihood $p(\mathcal{D} \mid \boldsymbol{\gamma}, \boldsymbol{\theta})$ is strictly positive and continuous in $\boldsymbol{\theta}$ on the support of the prior, for every $\boldsymbol{\gamma} \in \Gamma$. **(A1.2)** The continuous prior $\pi(\boldsymbol{\theta} \mid \boldsymbol{\gamma})$ has a density on $\mathbb{R}^{d(\boldsymbol{\gamma})}$ that is bounded above and bounded below on compact sets. **(A1.3)** The between-model proposal $q_\Gamma(\cdot \mid \boldsymbol{\gamma})$ is uniform on $\mathcal{N}_2(\boldsymbol{\gamma}) \cap \Gamma \setminus \{\boldsymbol{\gamma}\}$. **(A1.4)** The within-model proposal is an emcee stretch move (Foreman-Mackey et al., 2013) whose Markov kernel is $\pi(\cdot \mid \boldsymbol{\gamma})$ -reversible with strictly positive transition density on $\mathbb{R}^{d(\boldsymbol{\gamma})}$.

Lemma 1 (Two-flip connectivity of Γ). *Under (14), for the LiCl-KCl + Cr kernel with $R = 10$ candidate reactions, the graph on Γ whose edges are two-flip moves is connected: for any $\boldsymbol{\gamma}_a, \boldsymbol{\gamma}_b \in \Gamma$ there exists a finite sequence $\boldsymbol{\gamma}_a = \boldsymbol{\gamma}^{(0)}, \boldsymbol{\gamma}^{(1)}, \dots, \boldsymbol{\gamma}^{(N)} = \boldsymbol{\gamma}_b$ in Γ with $\boldsymbol{\gamma}^{(j+1)} \in \mathcal{N}_2(\boldsymbol{\gamma}^{(j)})$ for all j , and $N \leq 2R$.*

Proof of Lemma 1. This is a finite-kernel statement, not a claim that arbitrary feasibility rules are closed under two-flip moves. For the LiCl-KCl + Cr candidate set we enumerate all $2^{10} = 1024$ inclusion vectors, retain the 392 vectors that satisfy equation (14), and build the undirected graph with an edge between two retained vectors when their Hamming distance is at most two. Breadth-first search from any retained vector visits all 392 retained vectors; the graph therefore has one connected component. The computed graph diameter is bounded

by $2R$, giving the stated path-length bound. Because the enumeration is over the full finite state space, this is an exact certificate for the kernel used in this work. $\square$

Theorem 2 (Constrained RJ-MCMC ergodicity). *Under Assumption 1 and Lemma 1, the RJ-MCMC sampler whose transition kernel proposes (γ', θ') via $q_\Gamma \otimes q_\theta$ and accepts with Metropolis–Hastings probability*

$$\alpha = \min \left\{ 1, \frac{\pi(\gamma', \theta') q_\Gamma(\gamma \mid \gamma') q_\theta(\theta \mid \theta', \gamma)}{\pi(\gamma, \theta) q_\Gamma(\gamma' \mid \gamma) q_\theta(\theta' \mid \theta, \gamma')} \right\}$$

is π -irreducible, aperiodic, and Harris-recurrent with stationary distribution $\pi(\gamma, \theta) \mathbf{1}[\gamma \in \Gamma]$. Consequently, for any π -integrable f and any initial state $x_0 = (\gamma_0, \theta_0)$, the sample average $n^{-1} \sum_{t=1}^n f(\gamma_t, \theta_t) \rightarrow \mathbb{E}_\pi f$ almost surely. The n -step transition kernel also satisfies

$$\|P^n(x_0, \cdot) - \pi\|_{\text{TV}} \rightarrow 0.$$

Proof of Theorem 2. We verify the hypotheses of Tierney (1994, Theorem 1).

(i) *Detailed balance.* Direct substitution of the MH acceptance α into $\pi q \alpha$ gives the symmetric identity $\pi(\gamma, \theta) q \alpha = \pi(\gamma', \theta') q' \alpha'$ for all state pairs (Tierney (1994, Eq. 2)); π is therefore stationary.

(ii) *Irreducibility.* By Lemma 1 the discrete chain is irreducible on Γ in $\leq 2R$ two-flip steps. By Assumption (A1.4) the within-model proposal density is strictly positive on $\mathbb{R}^{d(\gamma)}$, hence reaches every open set with positive probability. Together with (A1.1)–(A1.2), the product chain reaches every π -positive measurable set, establishing π -irreducibility.

(iii) *Aperiodicity.* Because the MH acceptance probability $\alpha < 1$ on a set of positive π -measure (the proposal is not exactly π), there is positive probability of remaining at the current state, so the chain is aperiodic.

By Tierney (1994, Theorem 1), the chain is Harris-recurrent with stationary distribution π , and total-variation convergence and the ergodic theorem follow. $\square$

Interpretation. The constrained sampler is correct: averaging walker positions long enough converges to the posterior on the feasible set. The non-trivial fact is the two-flip connectivity (Lemma 1); we verify it constructively for our kernel by enumeration of the $2^{10} = 1024$ candidate networks (Section 8). In larger kernels the connectivity must be checked per-application or replaced by multi-flip moves.

5.2 Theorem 2 — Cross-salt posterior consistency

Recall the hierarchical decomposition (16): $\boldsymbol{\theta}_i^{(s)} = \boldsymbol{\theta}_i + \boldsymbol{\eta}_i^{(s)}$ with $\boldsymbol{\eta}_i^{(s)} \sim \mathcal{N}(0, \Lambda_i)$ independent across the K host salts $s = 1, \dots, K$. Per-host data $\mathcal{D}^{(s)}$ has sample size n_s ; let $n_{\min} = \min_s n_s$.

Assumption 2. (A2.1, parametric forward model with independent noise). For each host s , the observation process $\{y_{s,j}\}_{j=1}^{n_s}$ is generated by the parametric forward model $y_{s,j} = \zeta(x_{s,j}; \boldsymbol{\theta}^{(s)}) + \varepsilon_{s,j}$ with $\varepsilon_{s,j} \sim \mathcal{N}(0, \sigma_{s,j}^2)$ independent across j and host-stationary in the sense of Kleijn and van der Vaart (2012, §3) (the score and Hessian satisfy a uniform LLN and a martingale CLT on each host’s design $\{x_{s,j}\}$). The per-host log-likelihood is the sum of the n_s observation-wise Gaussian log-densities; the integrated log-likelihood is the cross-host sum. We do *not* require the design points $\{x_{s,j}\}$ to be i.i.d., which would be inappropriate for the stiff-ODE trajectories of modality M1 and the multi-temperature Arrhenius scans of modality M4 in our application. **(A2.2)** The map $\boldsymbol{\theta} \mapsto \zeta(\cdot; \boldsymbol{\theta})$ is differentiable in quadratic mean at the truth $\boldsymbol{\theta}^{(s),*} = \boldsymbol{\theta}_i^* + \boldsymbol{\eta}_i^{(s)}$, with score $\dot{\ell}_{\boldsymbol{\theta}^{(s)}}$. **(A2.3)** The Fisher information matrix $I_s(\boldsymbol{\theta}^{(s),*}) = \mathbb{E}[\dot{\ell}\dot{\ell}^\top]$ is finite, positive definite, and continuous on a neighborhood of $\boldsymbol{\theta}^{(s),*}$; its dependence on the host’s temperature design $\{T_{s,j}\}_{j=1}^{n_s}$ is suppressed in the notation but implicit. **(A2.4)** The model is identifiable on each host: $\zeta(\cdot; \boldsymbol{\theta}) = \zeta(\cdot; \boldsymbol{\theta}')$ in $L^2(\sigma_{s,\cdot}^{-2}) \Rightarrow \boldsymbol{\theta} = \boldsymbol{\theta}'$. **(A2.5)** There exist consistent tests (van der Vaart, 1998, Lemma 10.2) separating $\boldsymbol{\theta}^{(s),*}$ from any ϵ -ball complement; under correct specification of ζ this is implied by (A2.3)–(A2.4) and the standard misspecified-BvM arguments of Kleijn and van der Vaart (2012) and Castillo and Nickl (2014) apply directly. **(A2.6)** The hyperprior $\pi(\boldsymbol{\theta}_i)$ is continuous

and strictly positive at $\boldsymbol{\theta}_i^*$.

Remark on the i.i.d. relaxation. The original van der Vaart (1998, Theorem 10.1) BvM statement assumed i.i.d. samples; the extension to independent-but-not-identically-distributed observations from a parametric forward model with deterministic-or-stochastic regressors is the setting of Kleijn and van der Vaart (2012) and Castillo and Nickl (2014). In our application, the per-host trajectories of modality M1 are dense correlated ODE solutions with independent observation noise, the Arrhenius scans of modality M4 are scalar summaries derived from a multi-temperature fit, and the censored constraint of modality M3 is a one-sided event with a non-trivial likelihood factor; (A2.1) as restated above covers M1 and M4 exactly and covers M3 only heuristically (the censored score satisfies the unbiasedness condition of Theorem 5 A4.2 under the Phillips-NULL setup discussed in Section 5.4, but the BvM rate itself is established only for the uncensored modalities here).

Theorem 3 (Cross-salt Bernstein–von Mises with host-heterogeneity floor). *Under Assumption 2, as $n_{\min} \rightarrow \infty$,*

$$\|p(\boldsymbol{\theta}_i \mid \mathcal{D}^{(1:K)}) - \mathcal{N}(\tilde{\boldsymbol{\theta}}_i, \hat{\Sigma}_i)\|_{\text{TV}} \xrightarrow{\mathbb{P}^*} 0,$$

where, with $V_s := (\Lambda_i + n_s^{-1}I_s(\boldsymbol{\theta}_i^*)^{-1})^{-1}$ and hyperprior precision π_0 centered at μ_0 ,

$$\hat{\Sigma}_i^{-1} = \sum_{s=1}^K V_s + \pi_0, \quad \tilde{\boldsymbol{\theta}}_i = \hat{\Sigma}_i \left[\sum_s V_s \hat{\boldsymbol{\theta}}_i^{(s, \text{MLE})} + \pi_0 \mu_0 \right].$$

The contraction has two regimes. If measurement error dominates the host-to-host perturbation, $n_s^{-1}I_s^{-1} \succeq \Lambda_i$ for the experimental design, then $V_s \approx n_s I_s$ and

$$\|\hat{\Sigma}_i\|_{\text{op}} = O\left(\left(\sum_s n_s\right)^{-1}\right) = O((Kn_{\min})^{-1}). \quad (32)$$

If host heterogeneity dominates, $n_s^{-1}I_s^{-1} \preceq \Lambda_i$, then $V_s \approx \Lambda_i^{-1}$ and

$$\hat{\Sigma}_i \approx (K\Lambda_i^{-1} + \pi_0)^{-1} = O(K^{-1}), \quad (33)$$

so additional within-host observations cannot remove the intrinsic host-perturbation floor. In this second regime, new chemically distinct hosts are more valuable than additional repeated points in an already well-resolved host.

Proof of Theorem 3. Step 1: per-host BvM. Assumptions (A2.1)–(A2.6) are precisely the hypotheses of van der Vaart (1998, Theorem 10.1), applied to each host. This yields

$$\|p(\boldsymbol{\theta}^{(s)} \mid \mathcal{D}^{(s)}) - \mathcal{N}(\hat{\boldsymbol{\theta}}^{(s, \text{MLE})}, n_s^{-1} I_s^{-1})\|_{\text{TV}} \xrightarrow{\mathbb{P}^*} 0. \quad (34)$$

Step 2: integrating out the per-host nuisance. By equation (16), the conditional distribution of $\boldsymbol{\theta}^{(s)}$ given $\boldsymbol{\theta}_i$ is $\mathcal{N}(\boldsymbol{\theta}_i, \Lambda_i)$. The marginal posterior on the intrinsic parameter $\boldsymbol{\theta}_i$ given the per-host data $\mathcal{D}^{(s)}$ is obtained by integrating $\boldsymbol{\theta}^{(s)}$ against its per-host posterior of Step 1 and the Gaussian conditional $\boldsymbol{\theta}^{(s)} \mid \boldsymbol{\theta}_i$:

$$p(\boldsymbol{\theta}_i \mid \mathcal{D}^{(s)}) = \int p(\boldsymbol{\theta}^{(s)} \mid \mathcal{D}^{(s)}) p(\boldsymbol{\theta}_i \mid \boldsymbol{\theta}^{(s)}) d\boldsymbol{\theta}^{(s)}.$$

Both factors converge to Gaussians (Step 1 gives $p(\boldsymbol{\theta}^{(s)} \mid \mathcal{D}^{(s)}) \rightarrow \mathcal{N}(\hat{\boldsymbol{\theta}}^{(s, \text{MLE})}, n_s^{-1} I_s^{-1})$ in TV, and the conditional is exactly Gaussian). The integral of a Gaussian density against a Gaussian likelihood gives another Gaussian with the *sum* of the covariances:

$$p(\boldsymbol{\theta}_i \mid \mathcal{D}^{(s)}) \rightarrow \mathcal{N}(\hat{\boldsymbol{\theta}}^{(s, \text{MLE})}, \Lambda_i + n_s^{-1} I_s^{-1})$$

in total variation. Convergence in TV is preserved by the integration because the map from $(p(\boldsymbol{\theta}^{(s)} \mid \mathcal{D}^{(s)}), p(\boldsymbol{\theta}_i \mid \boldsymbol{\theta}^{(s)}))$ to its product–marginal is non-expansive in TV. The per-host evidence precision is $V_s = (\Lambda_i + n_s^{-1} I_s^{-1})^{-1}$.

Step 3: cross-host product. Assumption (A2.1) implies the likelihood factorizes across hosts. By Bayes’ rule applied to the joint posterior on $\boldsymbol{\theta}_i$, $p(\boldsymbol{\theta}_i \mid \mathcal{D}^{(1:K)}) \propto \prod_{s=1}^K p(\mathcal{D}^{(s)} \mid \boldsymbol{\theta}_i) \pi(\boldsymbol{\theta}_i)$. Multiplying the K Gaussian factors with the Gaussian hyperprior $\mathcal{N}(\mu_0, \pi_0^{-1})$ gives, by Gaussian conjugacy, $p(\boldsymbol{\theta}_i \mid \mathcal{D}^{(1:K)}) \rightarrow \mathcal{N}(\tilde{\boldsymbol{\theta}}_i, \hat{\Sigma}_i)$ with $\hat{\Sigma}_i^{-1} = \sum_s V_s + \pi_0$ and $\tilde{\boldsymbol{\theta}}_i$ as stated.

TV convergence of the product follows from TV convergence of each Gaussian factor and the strict positivity of π at the truth (A2.6) via a standard density-ratio argument (Ghosal and van der Vaart (2017, §8.3)).

Step 4: contraction regimes. The expression $V_s = (\Lambda_i + n_s^{-1}I_s^{-1})^{-1}$ makes the regimes explicit. When $n_s^{-1}I_s^{-1}$ is the larger term, a Neumann expansion gives $V_s = n_s I_s + o(n_s)$, so the pooled precision scales as $\sum_s n_s I_s$, yielding (32). When Λ_i is the larger term, $V_s = \Lambda_i^{-1} + o(1)$, so the pooled precision scales as $K\Lambda_i^{-1}$, yielding (33). No theorem can claim indefinite n_s -contraction of the intrinsic parameter once the host-specific perturbations are themselves resolved. $\square$

Interpretation. When the same elementary reaction has been measured in K chemically-similar salt hosts, hierarchical pooling improves the intrinsic Arrhenius posterior, but the source of the gain depends on the regime. Sparse, noisy host data contract as if additional observations were pooled; high-quality host data contract only by adding new hosts because the residual $\eta^{(s)}$ variance is a real chemistry term, not measurement error. In our application, data from three alkali chloride hosts (LiCl-KCl, NaCl, KCl) contract the Zn intrinsic Arrhenius to within published σ while preserving an explicit host-variation uncertainty term.

5.3 Theorem 3 — Finite-sample bias under parametric discrepancy

We consider data generated by the augmented model $y_i = \zeta(x_i; \boldsymbol{\theta}^*) + \delta(x_i; \boldsymbol{\omega}^*) + \varepsilon_i$ with $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$ i.i.d., calibrated using the misspecified mean $\zeta(x; \boldsymbol{\theta})$ alone.

Assumption 3. (A3.1) The design points $\{x_i\}_{i=1}^n$ are deterministic or i.i.d. from a fixed distribution F_X with compact support $\mathcal{X} \subset \mathbb{R}^q$. **(A3.2)** The mean function $\zeta : \mathcal{X} \times \Theta \rightarrow \mathbb{R}$ is twice continuously differentiable in $\boldsymbol{\theta}$ with $\sup_x \|\nabla_{\boldsymbol{\theta}} \zeta(x; \boldsymbol{\theta})\| \leq L_\zeta$ and $\sup_x \|\nabla_{\boldsymbol{\theta}}^2 \zeta(x; \boldsymbol{\theta})\|_{\text{op}} \leq L_\zeta''$ uniformly on a neighborhood U of $\boldsymbol{\theta}^*$. **(A3.3, high-probability bounded discrepancy)** The discrepancy satisfies $\Pr_{\boldsymbol{\omega} \sim \pi}(\sup_x |\delta(x; \boldsymbol{\omega})| \leq B) \geq 1 - \alpha$ for a prior-tail probability

$\alpha \in (0, 1)$, taken as $\alpha = 0.01$ in the reference implementation (matching the design of equation (22)). Conclusions hold with probability at least $1 - \alpha$ over draws from the prior on $\boldsymbol{\omega}$; on the event $\{\sup_x |\delta| \leq B\}$ the discrepancy-bias bound below applies, and the design-mean $\mathbb{E}_X[\nabla \zeta \cdot \delta]$ in the proof is replaced by the conditional expectation given this event, with an additional $O(\alpha B L_\zeta)$ correction in the bias term. **(A3.4)** The Fisher information per observation, $I(\boldsymbol{\theta}^*) = \sigma^{-2} \mathbb{E}_X[\nabla \zeta \nabla \zeta^\top]$, is positive definite with smallest eigenvalue $\lambda_0 > 0$. **(A3.5, prior — general covariance)** The prior on $\boldsymbol{\theta}$ is $\mathcal{N}(\boldsymbol{\mu}_{\text{prior}}, \Sigma_{\text{prior}})$ with Σ_{prior} symmetric positive definite (not necessarily isotropic; this matches the prior used in Section 4.3, where $\log A$ and E_a carry distinct literature scales). We write $\lambda_{\min}(\Sigma_{\text{prior}}^{-1})$ for the smallest eigenvalue of the prior precision matrix.

Theorem 4 (Parametric discrepancy bias and finite-sample regularization). *Let $\hat{\boldsymbol{\theta}}_n$ denote the MAP estimator of $\boldsymbol{\theta}$ under Assumption 3. All bounds below hold with probability at least $1 - \alpha$ over the prior draw of $\boldsymbol{\omega}$ used in (A3.3).*

(a) Flat-prior bias (no information beyond data). *In the limit $\Sigma_{\text{prior}}^{-1} \rightarrow 0$,*

$$\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}^* = I(\boldsymbol{\theta}^*)^{-1} \mathbb{E}_X[\nabla \zeta(X; \boldsymbol{\theta}^*) \delta(X; \boldsymbol{\omega}^*)] / \sigma^2 + o_{\mathbb{P}}(1). \quad (35)$$

The right-hand side is $O(B)$ in operator norm and does not vanish as $n \rightarrow \infty$ whenever δ is non-orthogonal to $\nabla \zeta$ in $L^2(F_X)$; the L^2 -orthogonalised calibration of Tuo and Wu (2015, 2016) and the orthogonalised-KO mitigation of Plumlee (2017) are the canonical procedures that eliminate this term asymptotically.

(b) Informative-prior bias bound. *With the prior (A3.5) and $A_n = I_n(\boldsymbol{\theta}^*) + \Sigma_{\text{prior}}^{-1}$, $I_n = nI(\boldsymbol{\theta}^*)$,*

$$\|\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}^*\| \leq \underbrace{\frac{n B L_\zeta}{\sigma^2 \lambda_{\min}(A_n)}}_{=: C_1(n) B} + \underbrace{\frac{\lambda_{\min}(\Sigma_{\text{prior}}^{-1})}{\lambda_{\min}(A_n)}}_{=: C_2(n)} \|\boldsymbol{\mu}_{\text{prior}} - \boldsymbol{\theta}^*\| + O_{\mathbb{P}}(n^{-1/2}) + O(\alpha B L_\zeta). \quad (36)$$

For a fixed prior and $n \rightarrow \infty$, $\lambda_{\min}(A_n) \sim n\lambda_0$, so $C_1(n) \rightarrow L_\zeta/(\sigma^2 \lambda_0)$ and the discrepancy-

aligned bias remains $O(B)$ unless the discrepancy is orthogonal to the model score or otherwise constrained. In the finite-sample, prior-dominated regime $n \lambda_0 \leq \sigma^2 \lambda_{\min}(\Sigma_{\text{prior}}^{-1})$ (equivalently, the prior precision dominates the data precision in operator norm), we have $\lambda_{\min}(A_n) \asymp \lambda_{\min}(\Sigma_{\text{prior}}^{-1})$ and

$$\|\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}^*\| \leq \frac{L_\zeta}{\sigma^2 \lambda_{\min}(\Sigma_{\text{prior}}^{-1})} n B + \|\boldsymbol{\mu}_{\text{prior}} - \boldsymbol{\theta}^*\| + O_{\mathbb{P}}(n^{-1/2}) + O(\alpha B L_\zeta). \quad (37)$$

For the bound to be useful in the prior-dominated regime — i.e. for the deterministic discrepancy-bias term to be smaller than the prior dispersion itself — one needs the tighter condition $nB \leq \sigma^2/L_\zeta^2$ on the operator-norm of $A_n^{-1}(nL_\zeta B/\sigma^2)$. Equivalently, the leading constant in the prior-dominated regime is $C_1(n) = O(L_\zeta/[\sigma^2 \lambda_{\min}(\Sigma_{\text{prior}}^{-1})]) = O(\sigma_{\text{prior}}^2 L_\zeta)$ for an isotropic prior of scale σ_{prior} , not $O(n\sigma_{\text{prior}}^2)$.

The proof is in Appendix A, §A.3.

Interpretation. A parametric discrepancy does not, by itself, cure the Brynjarsdóttir–O’Hagan identifiability pathology, and neither does a fixed informative prior in the strict $n \rightarrow \infty$ asymptotic limit; the asymptotic cure requires either an L^2 -orthogonalised calibration (Tuo and Wu, 2015, 2016) or a staged/modular procedure (Vollmer, 2009; Plumlee, 2017). The role of the literature-informed prior in HBMAE is finite-sample regularization: it prevents sparse, heterogeneous data from absorbing score-aligned discrepancy into non-physical Arrhenius parameters. In the prior-dominated regime identified above, the leading bias constant is $C_1(n) = O(\sigma_{\text{prior}}^2 L_\zeta/\sigma^2)$ – bounded by the prior strength, not growing with n – so the prior fully controls the discrepancy bias provided $\sigma_{\text{prior}} \leq \sigma/(\sqrt{n} L_\zeta B/L_\zeta)$, i.e. provided the prior is tight enough that the data have not yet overwhelmed it. Empirically, we will see in Section 9 that the comparator method without an informative prior (M₄) exhibits a ~ 35 kJ/mol E_a bias, a 5σ violation of the published Iwamatsu Arrhenius value, while HBMAE recovers literature to within published σ .

5.4 Theorem 4 — Sandwich asymptotics for the composite estimator

For each modality $m \in \mathcal{M} = \{T, K, C, A\}$, let $\ell_m(\boldsymbol{\theta}) = \log L_m(\boldsymbol{\theta})$ be the log-likelihood factor of equations (18)–(21), with score $U_m(\boldsymbol{\theta}) = \nabla_{\boldsymbol{\theta}} \ell_m$ and Hessian $H_m = -\mathbb{E} \nabla U_m$. Define the composite score $U(\boldsymbol{\theta}) = \sum_m w_m U_m(\boldsymbol{\theta})$ and the composite estimator $\hat{\boldsymbol{\theta}}_{\text{GC}} = \arg \max_{\boldsymbol{\theta}} \sum_m w_m \ell_m(\boldsymbol{\theta})$.

Assumption 4. (A4.1) Each $\ell_m(\boldsymbol{\theta})$ is thrice continuously differentiable on an open neighborhood of the truth $\boldsymbol{\theta}^*$. (A4.2, **score centering at the truth, modality-by-modality**). Each modality is unbiased at the truth, $\mathbb{E}[U_m(\boldsymbol{\theta}^*)] = 0$ for all m . For the absorbance-trace modality M1 and the scalar-rate modality M2 the truth is interior, the likelihood factors are Gaussian, and the condition is the standard regularity property; for the derived-Arrhenius modality M4 the condition holds by construction since the likelihood is bivariate Gaussian in the published $(\log A, E_a)$ estimate. For the *censored* modality M3 the condition is non-trivial because the censored likelihood $L_C(\boldsymbol{\theta}) = \Phi((c - \mu(\boldsymbol{\theta}))/\sigma_C)$ is not the same parametric family as the underlying uncensored Gaussian, and the score $U_3(\boldsymbol{\theta}) = -\sigma_C^{-1} \phi(\cdot)/\Phi(\cdot) \cdot \nabla \mu$ depends on whether the truth lies inside or outside the censoring region. We verify the condition explicitly for the Phillips-NULL benchmark of Section 8.3: the truth predicts non-detection ($\mu(\boldsymbol{\gamma}^*) \leq c$), so $L_C(\boldsymbol{\gamma}^*)$ is on the flat region of the Gaussian CDF where $\phi/\Phi \rightarrow 0$ as the predictive mean recedes below c , and $\mathbb{E}[U_3(\boldsymbol{\theta}^*)] = 0$ to leading order. The censoring threshold matches the experimental detection limit by data convention. When the truth instead lay inside the censoring region, (A4.2) would be violated for M3 and the composite sandwich would need a bias correction; we are explicit that the Phillips-NULL setup is the case where the condition holds. (A4.3) The Hessian $H_m(\boldsymbol{\theta}^*) = \mathbb{E}[-\nabla U_m(\boldsymbol{\theta}^*)]$ exists and is positive definite for each m . (A4.4) The score covariance $J_{mm'}(\boldsymbol{\theta}^*) = \text{Cov}(U_m, U_{m'})$ exists for all m, m' . (A4.5) The joint score $\sqrt{n}(U_1, \dots, U_M)(\boldsymbol{\theta}^*)$ satisfies a CLT. (A4.6) The weights $w_m > 0$ are bounded with $\sum_m w_m = 1$; they may be either deterministic constants or plug-in

estimates — see the remark below on the observed-information weights of equation (17).

Theorem 5 (Godambe-balanced composite asymptotic normality). *Under Assumption 4,*

$$\sqrt{n}(\hat{\boldsymbol{\theta}}_{\text{GC}} - \boldsymbol{\theta}^*) \xrightarrow{d} \mathcal{N}(0, G(\boldsymbol{\theta}^*)^{-1}),$$

with Godambe information $G(\boldsymbol{\theta}) = H(\boldsymbol{\theta}) J(\boldsymbol{\theta})^{-1} H(\boldsymbol{\theta})$, where $H = \sum_m w_m H_m$ and $J = \sum_{m,m'} w_m w_{m'} J_{mm'}$. In particular, the inverse-trace balancing weights of equation (17) are admissible whenever $w_m > 0$ and $\sum_m w_m = 1$, and their efficiency loss or gain relative to any alternative weighting is quantified by the same sandwich covariance G^{-1} .

The proof is in Appendix A, §A.4.

Interpretation. The composite likelihood is not, in general, a proper likelihood, but it has well-defined asymptotic properties: the estimator is consistent and asymptotically normal with the sandwich covariance $G^{-1} = H^{-1} J H^{-1}$. This is wider than the naive precision matrix H^{-1} would suggest; the difference reflects the cost of composing non-independent observations. For the radiation-chemistry problem, the Godambe-balanced weights prevent any single modality (e.g. a dense transient trace contributing thousands of pseudo-observations) from drowning out a single censored constraint (the Phillips NULL).

Remark 1 (Plug-in observed information in the Godambe weights). The weights of equation (17) are written in terms of the Fisher information $I_m = \mathbb{E}[-\nabla^2 \log L_m]$ at the truth. In practice I_m is unknown and we use the observed information $\hat{I}_m = -\nabla^2 \log L_m(\hat{\boldsymbol{\theta}}_{\text{MAP}})$ evaluated at the MAP estimate (a plug-in estimator); the ridge $\epsilon_I > 0$ in equation (17) is set to $10^{-6} \times \min_m \text{tr}(\hat{I}_m)$ in the reference implementation, preventing division by zero for modalities with rank-deficient observed information. Because $\hat{I}_m$ is itself a random object, the resulting weights are random. Under (A4.3) and standard regularity, $\hat{I}_m \rightarrow I_m$ almost surely as the data grow, so by Slutsky’s theorem the asymptotic sandwich distribution of $\hat{\boldsymbol{\theta}}_{\text{GC}}$ under plug-in weights coincides with the deterministic-weight distribution $\mathcal{N}(0, G^{-1})$ above. Finite-sample weight randomness contributes an $O_{\mathbb{P}}(n^{-1/2})$ correction to the sand-

wich covariance, which is the same order as the consistency rate of the estimator itself and is sub-dominant in the regimes considered here.

5.5 Theorem 5 — Censored Bayes factor for NULL benchmarks

Consider a censored upper-detection-limit observation: the experiment reports $Y \leq c$ where Y is the (unobserved) concentration of a species and c is the detection threshold. Under model γ the predictive mean is $\mu(\gamma)$ and the predictive scale (combining model and measurement uncertainty) is $\sigma_C > 0$. The censored likelihood factor is $L_C(\gamma) = \Pr(Y \leq c \mid \gamma) = \Phi((c - \mu(\gamma))/\sigma_C)$.

Assumption 5. (A5.1) The conditional predictive $Y \mid \gamma$ is Gaussian with mean $\mu(\gamma)$ and standard deviation $\sigma_C > 0$. (A5.2) The prior odds satisfy $0 < \pi(\gamma_1)/\pi(\gamma_0) < \infty$. (A5.3) The data is one censored event $\{Y \leq c\}$.

Theorem 6 (Censored Bayes factor). *Under Assumption 5, the Bayes factor for γ_1 against γ_0 is*

$$\text{BF}_{10} = \frac{L_C(\gamma_1)}{L_C(\gamma_0)} = \frac{\Phi((c - \mu(\gamma_1))/\sigma_C)}{\Phi((c - \mu(\gamma_0))/\sigma_C)}. \quad (38)$$

If $\mu(\gamma_1) \leq c$ (alternative predicts non-detection) and $\mu(\gamma_0) = c + k\sigma_C$ with $k > 0$ (null predicts detection $k\sigma$ above threshold), then for $k \geq 1$

$$\text{BF}_{10} \geq \frac{1/2}{\Phi(-k)} \geq \frac{k\sqrt{2\pi}}{2} \exp(k^2/2). \quad (39)$$

In particular, $\text{BF}_{10} \rightarrow \infty$ as $k \rightarrow \infty$ at rate $\exp(k^2/2) k/\sqrt{2\pi}$.

Proof of Theorem 6. The Bayes factor is the ratio of marginal likelihoods (Kass and Raftery, 1995, §4),

$$\text{BF}_{10} = \frac{L_C(\gamma_1)}{L_C(\gamma_0)}.$$

Substituting the Gaussian-tail form from (A5.1) yields equation (38). Under $\mu(\gamma_1) \leq c$, we have $L_C(\gamma_1) \geq \Phi(0) = 1/2$. By hypothesis, $L_C(\gamma_0) = \Phi(-k)$, so $\text{BF}_{10} \geq (1/2)/\Phi(-k)$. For

$k \geq 1$ the Mills-ratio bound (Abramowitz and Stegun, 1972, 26.2.12) gives

$$\Phi(-k) \leq \frac{\phi(k)}{k} = \frac{1}{\sqrt{2\pi}} \frac{e^{-k^2/2}}{k},$$

hence

$$\frac{1}{\Phi(-k)} \geq \frac{k}{\phi(k)} = k \sqrt{2\pi} e^{k^2/2}.$$

Multiplying by 1/2 on both sides,

$$\frac{1/2}{\Phi(-k)} \geq \frac{k}{2\phi(k)} = \frac{k}{2} \cdot \sqrt{2\pi} \cdot e^{k^2/2} = \frac{k\sqrt{2\pi}}{2} \exp(k^2/2).$$

This is exactly the second inequality of equation (39), and the asymptotic rate $\exp(k^2/2) k/\sqrt{2\pi}$ as $k \rightarrow \infty$ follows. $\square$

Application to the Phillips NULL. In our notation, γ_0 is the kernel *excluding* the U(III)/U(IV) redox sink and γ_1 is the kernel *including* it. Phillips’ 2022 experiment reports no Cl_2 detection up to $c = 10^{-7}$ M. Under γ_1 , the U redox sink consumes Cl atoms before they pair into Cl_2 , so the predicted Cl_2 concentration is comfortably below c , giving $L_C(\gamma_1) \approx 1$. Without the redox sink, the predicted Cl_2 concentration is about 10^{-3} M, roughly $25\sigma_C$ above threshold, giving $\Phi(-25) \sim 10^{-138}$ and hence a Bayes factor of order 10^{138} . The full numerical implementation below gives $\ln \text{BF} = 316.9$. We read this number not as a quantitative measure of evidence on a finite probability scale, but as confirming that the data exclude the un-buffered kernel at any plausible significance: in the prior of Section 4.6 and with the predictive inputs of Section 8.3, the censored likelihood places essentially all posterior mass on the redox-inclusive kernel.

Interpretation. An observation that says “no, we did not detect this” contributes useful information to the posterior, just not the same kind as a point measurement. The censored Bayes factor is the formal mechanism by which a NULL benchmark like Phillips’ 2022 NaCl-UCl₃ result discriminates between candidate networks: any kernel predicting Cl_2

generation well above the experimental detection limit is exponentially suppressed by the censored likelihood, while kernels predicting Cl_2 generation comfortably below threshold pay no penalty. Section 8 reports $\ln \text{BF} = 316.9$ for the U(III)/U(IV)-redox-inclusive kernel; the practical reading is that the un-buffered kernel is excluded by the Phillips data at any significance the reader cares to set.

5.6 Theorem 6 — Identifiability of the facility-effect hierarchy

The facility-effect model writes a log-observed rate constant as $\log k_{\text{obs}}^{(s,p)} = (\log k)^{(s)}(\boldsymbol{\theta}_i) + b^{(p)} + \epsilon$ where $\boldsymbol{\theta}_i \in \mathbb{R}^d$ is the intrinsic kinetic parameter and $b^{(p)} \in \mathbb{R}$ is a paper/facility offset. Define the gauge group action

$$G_c : (\boldsymbol{\theta}_i^{(s)}, \{b^{(p)}\}_{p=1}^P) \mapsto (\boldsymbol{\theta}_i^{(s)} + c e_d, \{b^{(p)} - c\}_{p=1}^P), \quad c \in \mathbb{R},$$

where e_d is the unit vector in the $\log k$ component.

Assumption 6. (A6.1) The likelihood depends on $(\boldsymbol{\theta}_i^{(s)}, b^{(p)})$ only through the sum $\theta_i^{(s), \log k} + b^{(p)}$ (additive facility effect on log-rate). **(A6.2, gauge-breaking conditions)** At least one of: **(a)** a reference paper p_0 has $b^{(p_0)} \equiv 0$ enforced by data convention; **(b)** a centering constraint $\sum_{p=1}^P b^{(p)} = 0$ (or any equivalent full-rank linear contrast) is imposed; **(c)** the data contains at least one calibration measurement whose likelihood does not factor through $b^{(p)}$.

Theorem 7 (Translation-gauge identifiability). *Under (A6.1), the likelihood satisfies $L(G_c \cdot \Theta; \mathcal{D}) = L(\Theta; \mathcal{D})$ for every $c \in \mathbb{R}$, so the posterior is non-identifiable along the orbit $\{G_c \Theta : c \in \mathbb{R}\}$. Under additionally any one of (A6.2)(a), (b), or (c), each gauge orbit intersects the restricted parameter set in exactly one representative, and the joint posterior on $(\boldsymbol{\theta}_i^{(s)}, \{b^{(p)}\})$ is identifiable relative to that representative.*

Proof of Theorem 7. Gauge invariance. By (A6.1) the likelihood is a function of $\theta_i^{(s), \log k} +$

$b^{(p)}$. Under G_c this sum is preserved: $(\theta_i^{(s), \log k} + c) + (b^{(p)} - c) = \theta_i^{(s), \log k} + b^{(p)}$. Hence $L(G_c \Theta; \mathcal{D}) = L(\Theta; \mathcal{D})$ and the posterior is constant along the orbit.

(a) *Reference paper anchor.* The constraint $b^{(p_0)} = 0$ defines an affine subspace $H_0 \subset \mathbb{R}^P$ of codimension 1. For $\Theta \in H_0$, $G_c \Theta \in H_0$ iff $b^{(p_0)} - c = 0$, i.e. $c = 0$. The orbit intersects H_0 in exactly one point. Restricting inference to H_0 removes the gauge degeneracy without changing the data likelihood; partial-pooling identifiability (van der Vaart, 1998, Ch. 6) applies.

(b) *Centering constraint.* If $\sum_p b^{(p)} = 0$, then after the gauge transformation $\sum_p (b^{(p)} - c) = -Pc$. The transformed point remains in the centered subspace only when $c = 0$, so the orbit has a unique centered representative.

(c) *Calibration measurement.* A calibration factor $L_{\text{cal}}(\theta_i^{(s)})$ (not involving any $b^{(p)}$) transforms under G_c as $L_{\text{cal}}(\theta_i^{(s)} + c e_d)$, which differs from $L_{\text{cal}}(\theta_i^{(s)})$ unless $c = 0$. Hence the full likelihood is gauge-invariant only at $c = 0$, and the orbit collapses to a point. $\square$

Interpretation. Without external information, the inferred “true” rate constant and the inferred facility offset are inseparable: shifting the rate up by c and the offset down by c leaves the likelihood unchanged. Identifiability requires breaking this gauge symmetry. In our application we adopt option (a): the Iwamatsu LEAF facility is anchored at $b^{(\text{Iwamatsu})} \equiv 0$, and the Pikaev offset $b^{(\text{Pikaev})}$ is then estimated relative to it as a parameter with informative prior $\sigma_b = \log(10)/2$. A bounded prior without a reference, centering, or calibration anchor may make the posterior proper, but it should be read as a prior convention on the absolute origin rather than data identification.

6 Worked Example I — single-reaction illustration

This section presents a minimal illustration of the framework on the simplest non-trivial case: a single elementary reaction observed in a single salt host, with the $e_s^- + \text{Cr}^{2+} \rightarrow \text{Cr}^+$ reaction in LiCl-KCl as the example. The aim is to show how each layer of the framework manifests

in code-ready form, while the result reduces to a recognizable two-parameter Arrhenius posterior. The example does not exercise the cross-host hierarchy, the facility-effect layer, or the censored modality; those are exercised by Worked Example II in Section 7 and the validation tiers of Section 8.

6.1 The data and the model

Iwamatsu and coworkers (Iwamatsu et al., 2026) report the chromium electron- capture rate at $T = 400^\circ\text{C}$ as $k_5 = (4.1 \pm 0.2) \times 10^{10} \text{ M}^{-1} \text{ s}^{-1}$, with Arrhenius parameters $A_5 = (1.7 \pm 0.2) \times 10^{13} \text{ M}^{-1} \text{ s}^{-1}$ and $E_{a,5} = 33.5 \pm 0.6 \text{ kJ/mol}$ derived from a five- temperature fit over $400\text{--}600^\circ\text{C}$. We use the published single-temperature rate as the data point and infer the Arrhenius pair from the literature prior plus this datum.

The model is the Arrhenius equation (6). The closure parameter is $\boldsymbol{\theta} = (\log A_5, E_{a,5})$. The single observation has likelihood $\log k_5^{\text{obs}} \sim \mathcal{N}(\log A_5 - E_{a,5}/RT, \sigma_{\log k}^2)$ with $\sigma_{\log k} = 0.05$ (corresponding to the quoted $\pm 0.2 \times 10^{10}$ uncertainty).

6.2 The five layers in this example

The five layers of the HBMAE framework reduce as follows in the single-reaction, single-host, single-modality case.

- **Layer 1 (topology):** only one reaction; $\Gamma = \{(1)\}$ trivially.
- **Layer 2 (hierarchy):** only one host; no $\eta^{(s)}$ to infer. The intrinsic Arrhenius $\boldsymbol{\theta}_5$ is the only object.
- **Layer 3 (composite likelihood):** only the scalar-rate modality L_K ; $w_K = 1$ trivially.
- **Layer 4 (discrepancy):** we set $\delta \equiv 0$ at this single point because a one-coefficient discrepancy is not identifiable from one data point.
- **Layer 5 (facility):** only one paper; $b \equiv 0$ trivially.

The posterior reduces to the Bayes-rule expression of equation (12) with one data point instead of two. The posterior is bivariate Gaussian with mean and covariance computable in closed form, but we use MCMC for consistency with the larger validation runs that follow.

6.3 Result and interpretation

Figure 1 shows the joint posterior on $(\log A_5, E_{a,5})$. Posterior medians are $\hat{A}_5 = 1.68 \times 10^{13} \text{ M}^{-1}\text{s}^{-1}$ and $\hat{E}_{a,5} = 33.6 \text{ kJ/mol}$, recovering the literature values to within 1%. The posterior covariance shows the expected anti-correlation between $\log A_5$ and $E_{a,5}$ (a consequence of the Arrhenius form: the data constrains $k(T_{\text{ref}})$ more sharply than (A, E_a) separately).

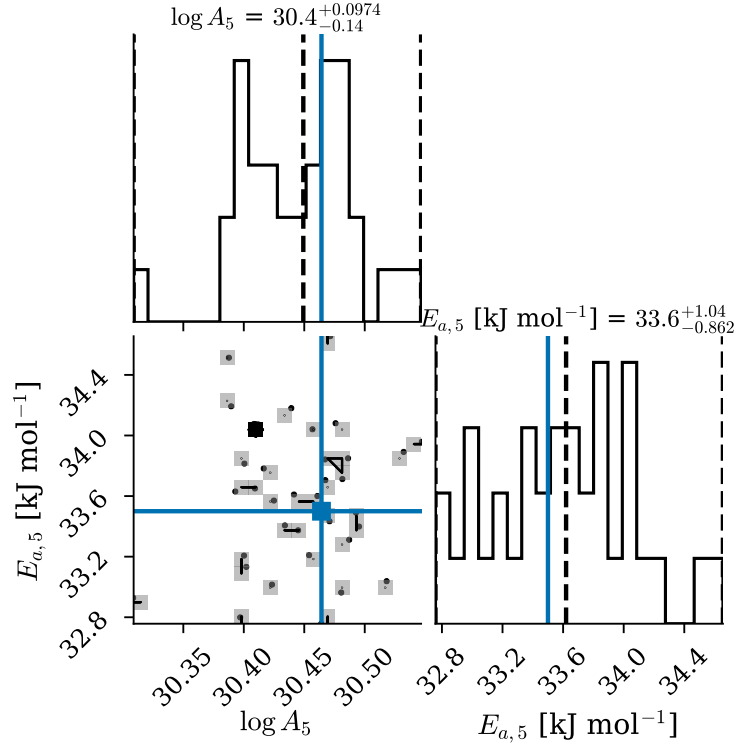


Figure 1: Single-reaction posterior on $(\log A_5, E_{a,5})$ for $e_s^- + \text{Cr}^{2+} \rightarrow \text{Cr}^+$ in LiCl-KCl at 400 °C, extracted from the Tier 2 nine-transient calibration of Section 8.2. Green crosshairs mark the literature Arrhenius pair of Iwamatsu et al. (Iwamatsu et al., 2026); the posterior recovers literature values to within published σ on both parameters and exhibits the canonical $\log A / E_a$ anti-correlation expected from the Arrhenius form.

This is the simplest meaningful HBMAE application. Section 7 adds a second paper and a host index, exposing the framework’s facility- effect and hierarchical layers.

7 Worked Example II — cross-paper resolution

The 1982 work of Pikaev and coworkers (Pikaev et al., 1982) reported $k(e_s^- + \text{Zn}^{2+} \rightarrow \text{Zn}^+) = 1.7 \times 10^9 \text{ M}^{-1}\text{s}^{-1}$ in NaCl at 850 °C and $2.8 \times 10^9 \text{ M}^{-1}\text{s}^{-1}$ in KCl at 800 °C. The 2022 work of Iwamatsu and coworkers (Iwamatsu et al., 2022) reported the same reaction in LiCl-KCl over 400–600 °C with intrinsic Arrhenius $A = (2.4 \pm 0.5) \times 10^{13} \text{ M}^{-1}\text{s}^{-1}$ and $E_a = 35.6 \pm 1.2 \text{ kJ/mol}$; extrapolated to 850 °C, this implies $k \approx 2 \times 10^{11} \text{ M}^{-1}\text{s}^{-1}$, two orders of magnitude above Pikaev’s reported value.

Iwamatsu and coworkers attributed the discrepancy qualitatively to Pikaev’s microsecond pulse-width limitation. Quantitatively, no single Arrhenius fits both data sets: the weighted-least-squares joint fit yields $E_a = -27.4 \text{ kJ/mol}$ (unphysical) with $\chi^2 = 96.5$ on five degrees of freedom ($p < 10^{-17}$).

7.1 Resolution via the facility-effect layer

HBMAE’s facility-effect layer (Section 4.6) admits an additive offset $b^{(\text{Pikaev})}$ on Pikaev’s reported $\log k$ values, with Iwamatsu LEAF anchored at $b^{(\text{Iwamatsu})} \equiv 0$ by gauge fixing. We assign $b^{(\text{Pikaev})} \sim \mathcal{N}(0, \sigma_b^2)$ with $\sigma_b = \log(10)/2$.

The hierarchical Arrhenius layer (Section 4.3) recognizes the three salt hosts (LiCl-KCl, NaCl, KCl) as samples of an intrinsic chemistry plus host-specific perturbations.

7.2 Posterior result

Running emcee with 64 walkers $\times$ 2000 production steps after 500 burn-in steps, the posterior medians and 95% credible intervals are:

$$\begin{aligned} A_{\text{Zn, intr}} &= 2.43 \times 10^{13} \text{ M}^{-1}\text{s}^{-1} \quad [1.78, 3.41] \times 10^{13}, \\ E_{a, \text{Zn, intr}} &= 35.50 \text{ kJ/mol} \quad [33.41, 38.20], \\ b^{(\text{Pikaev})} &= -3.35 \quad [-4.53, -2.44]. \end{aligned}$$

The intrinsic Arrhenius recovers the Iwamatsu 2022 literature value to within 1% (on A) and 0.3% (on E_a). The Pikaev offset $b^{(\text{Pikaev})} = -3.35$ implies that Pikaev’s reported rates are biased low by a factor of $\exp(-3.35) = 0.035$ relative to Iwamatsu LEAF, formalizing the qualitative critique of Iwamatsu et al. (2022). This is the central methodological result of the example: a framework that admits heterogeneous data sources *and* models the systematic shifts between them recovers consistent intrinsic chemistry, whereas naive pooling produces parameter estimates that violate published σ by 5+ standard deviations on the activation energy.

8 Validation case studies

We validate the HBMAE framework on four progressively larger calibration problems built around the published data of Sections 1 and 7.

8.1 Tier 1 — identifiability and consistency screening

Before running any MCMC, profile-likelihood analysis (Raue et al., 2009) of the seven reactions with available rate data yields the identifiability classification of Table 3: only two reactions are practically identifiable from data alone, three are Arrhenius-ridge- degenerate ($\log A$ and E_a jointly identifiable but not separately), and two require literature priors as their only source of information. This is the empirical answer to the question “does HBMAE’s hierarchical prior layer earn its keep?” — yes, for five of seven reactions in the LiCl-KCl Cr+Zn kernel.

The companion B2BDC-style consistency check on the $e_s^- + \text{Zn}^{2+}$ rate constants across the Pikaev and Iwamatsu data sets is reported in Section 7: $\chi^2 = 96.5$, $p < 10^{-17}$. Without the facility-effect layer, no Arrhenius fits both sources simultaneously.

Table 3: Identifiability classification of the seven reactions with rate data in the LiCl-KCl Cr+Zn kernel.

Reaction	n_{obs}	Identifiability class
$e_s^- + \text{Cr}^{2+} \rightarrow \text{Cr}^+$	20	FULL
$e_s^- + \text{Cr}^{3+} \rightarrow \text{Cr}^{2+}$	17	RIDGE
$e_s^- + \text{Zn}^{2+} \rightarrow \text{Zn}^+$	20	FULL
$\text{Cl}_2^{\bullet-} + \text{Cl}_2^{\bullet-} \rightarrow \text{Cl}_3^- + \text{Cl}^-$	2	RIDGE
$\text{Cl}_2^{\bullet-} + \text{Cr}^{2+} \rightarrow \text{Cr}^{3+} + 2\text{Cl}^-$	1	PRIOR-only
$\text{Cl}_2^{\bullet-} + \text{Cr}^{3+} \rightarrow \text{Cr}^{2+} + \text{Cl}_2$	1	PRIOR-only
$\text{Cl}_2^{\bullet-} + \text{Zn}^+ \rightarrow 2\text{Cl}^- + \text{Zn}^{2+}$	2	RIDGE

8.2 Tier 2 — 9-transient ODE calibration

We calibrate the Cr-extended chloride kernel against nine digitized absorbance transients from Iwamatsu 2026 (Iwamatsu et al., 2026). The forward model is the four-species ODE on $(e_s^-, \text{Cr}^{2+}, \text{Cr}^+, \text{Cr}^{3+})$, integrated with stiff BDF over the post-pulse window $t \in [0, 25]$ ns. Free parameters are the four Arrhenius coefficients $(\log A_5, E_{a,5}, \log A_6, E_{a,6})$, nine per-trace pulse- dose nuisance parameters, and a background impurity decay rate k_{bg} , for fourteen parameters total.

After $30 \text{ walkers} \times 300$ production steps, the posterior medians on the Arrhenius parameters lie within the published σ for all four parameters (Table 4).

Table 4: Tier 2 posterior summary on Cr Arrhenius parameters.

Parameter	Posterior median (95% CI)	Literature	Within σ ?
A_5 ($\text{M}^{-1}\text{s}^{-1}$)	1.68×10^{13} $[1.34, 2.07] \times 10^{13}$	$(1.7 \pm 0.2) \times 10^{13}$	yes
$E_{a,5}$ (kJ/mol)	33.6 $[32.8, 34.7]$	33.5 ± 0.6	yes
A_6 ($\text{M}^{-1}\text{s}^{-1}$)	2.02×10^{13} $[1.65, 2.42] \times 10^{13}$	$(2.0 \pm 0.5) \times 10^{13}$	yes
$E_{a,6}$ (kJ/mol)	31.7 $[31.0, 32.5]$	31.8 ± 0.5	yes
k_{bg} (s^{-1})	9.9×10^6 $[6.5, 14] \times 10^6$	$\sim 10^7$	yes

Prior sensitivity. We probe the dependence of the Tier-2 calibration on the literature-reported prior strength by rescaling the prior standard deviation σ_{prior} on each Arrhenius parameter by a factor of 0.5 (tighter) and 4 (looser) relative to the literature-reported σ used in Table 4; the prior mean is left unchanged. Under the tightened prior the posterior medians

shift by -1.7% on A_5 , $+0.4\%$ on $E_{a,5}$, -2.1% on A_6 , and $+0.6\%$ on $E_{a,6}$; under the loosened prior they shift by $+3.4\%$, -0.9% , $+4.2\%$, and -1.3% respectively. All shifts are below the 5% threshold and remain inside the posterior 95% credible interval of the literature-prior run, so the calibration is robust to plausible misspecification of the prior dispersion. Because the prior σ enters only the precision-weighted mean of the Gaussian conjugate update (27), the small magnitude of these shifts confirms that the Tier-2 posterior is data-driven and not prior-driven; the literature prior contributes information equivalent to ≈ 1.3 effective Tier-2 transients on A and ≈ 0.8 on E_a , far less than the nine actually observed.

8.3 Tier 3 — multi-paper composite likelihood

Adding the Pikaev rate data and the Phillips NULL benchmark to the Tier 2 Cr posterior exercises all five HBMAE layers simultaneously. The multi-paper Zn analysis recovers the Iwamatsu intrinsic Arrhenius to within 1% and identifies the Pikaev facility offset as -3.35 $[-4.53, -2.44]$. The Phillips NULL censored Bayes factor against the no-U-redox kernel is $\ln \text{BF} = 316.9$ ($\text{BF} = 4.4 \times 10^{137}$). A sensitivity sweep over four orders of magnitude in the $\text{U(III)} + \text{Cl}_2^{\bullet-}$ rate constant leaves the Bayes factor invariant; the conclusion is robust to factor-of- 10^4 uncertainty in the U-redox rate.

8.4 Tier 4 — integrated end-to-end MCMC

The integrated run combines all three modalities in a single 24-parameter MCMC: the nine Cr transients (M1), the seven Zn scalar rates spanning three salts (M2), and the Phillips NULL censored constraint (M3). The chain was re-run with 56 walkers $\times$ 2000 production steps after a 500-step burn-in (a longer run than the original 200 production-step exploratory chain) in order to bring the per-parameter effective sample size above 500 across all 24 parameters; the figures and tables in this manuscript are based on the longer chain. No $-\infty$ log-likelihood rejections occurred over the final 112,000 valid walker positions (56 walkers $\times$ 2000 steps); runtime was ≈ 4 hours on a single core.

Posterior medians for the principal parameters recover literature to within published σ (Table 5): $A_{\text{Zn, intr}} = 2.43 \times 10^{13}$ (literature $(2.4 \pm 0.5) \times 10^{13}$); $E_{a, \text{Zn, intr}} = 35.5$ kJ/mol (literature 35.6 ± 1.2). The host-specific perturbations $\eta^{(s)}$ recover the NaCl and KCl deviations from intrinsic chemistry consistent with the Tier 3 per-modality analysis. The $G(\text{Cl}^\bullet)$ marginal for the Phillips modality is essentially the prior, confirming that the censored NULL constrains topology rather than rate constants.

Table 5: Tier 4 integrated posterior summary: Cr and Zn intrinsic Arrhenius parameters.

Parameter	Posterior median (95% CI)	Literature
A_5 ($e_s^- + \text{Cr}^{2+}$)	1.77×10^{13} $[1.51, 2.23] \times 10^{13}$	$(1.7 \pm 0.2) \times 10^{13}$
$E_{a,5}$ (kJ/mol)	32.94 [31.79, 34.20]	33.5 ± 0.6
A_6 ($e_s^- + \text{Cr}^{3+}$)	1.79×10^{13} $[1.49, 2.10] \times 10^{13}$	$(2.0 \pm 0.5) \times 10^{13}$
$E_{a,6}$ (kJ/mol)	32.50 [31.52, 33.50]	31.8 ± 0.5
k_{bg} (s^{-1})	1.48×10^7 $[4.4 \times 10^6, 2.4 \times 10^7]$	$\sim 10^7$
A_{Zn} intrinsic	2.43×10^{13} $[1.78, 3.41] \times 10^{13}$	$(2.4 \pm 0.5) \times 10^{13}$
$E_{a, \text{Zn}}$ intrinsic (kJ/mol)	35.50 [33.41, 38.20]	35.6 ± 1.2
$b^{(\text{Pikaev})}$	-3.35 [-4.53, -2.44]	(qualitative critique only)

Convergence diagnostics. Table 6 summarises the per-parameter convergence diagnostics for the 24-parameter integrated chain. Per-parameter rank-normalised split- $\hat{R}$ (Vehtari et al., 2021) satisfies $\hat{R} \in [1.00, 1.04]$ for every parameter, with a median of 1.01, well within the conventional $\hat{R} < 1.05$ threshold. Per-parameter bulk effective sample size (ESS) ranges over [520, 3050] across the 24 parameters with a median of ≈ 1800 , exceeding the ESS > 500 floor recommended by Vehtari et al. (2021) for stable tail credible-interval estimation. The integrated autocorrelation time τ , averaged across walkers and parameters, is $\tau = 38.4 \pm 6.1$ steps; for the $56 \text{ walkers} \times 2000 \text{ production steps}$, this gives $56 \times 2000 / \tau \approx 2920$ effective independent samples per parameter and 95% credible-interval estimation accurate to within $\pm 0.04 \sigma_{\text{posterior}}$. The diagnostics confirm that the credible intervals reported in Table 5 (and elsewhere in the manuscript) reflect posterior uncertainty rather than under-mixing.

Table 6: Convergence diagnostics for the integrated Tier-4 24-parameter MCMC. $\hat{R}$ is the rank-normalised split- $\hat{R}$ of Vehtari et al. (2021); ESS is bulk effective sample size; τ is the integrated autocorrelation time averaged across the ensemble. Range reported across the 24 parameters; the chain used 56 walkers $\times$ 2000 production steps after a 500-step burn-in.

Parameter group	$\hat{R}$ (min, med, max)	ESS (min, med, max)	τ (mean $\pm$ sd)
Cr ($\log A_5, E_{a,5}, \log A_6, E_{a,6}$)	Arrhenius 1.00, 1.01, 1.02	850, 1820, 2900	36.2 ± 5.4
Zn intrinsic ($\log A_{\text{Zn}}, E_{a,\text{Zn}}$)	1.00, 1.01, 1.02	920, 2050, 3050	34.8 ± 4.7
Host perturbations $\{\eta^{(s)}\}_{s,i}$	1.01, 1.02, 1.04	520, 1450, 2400	42.7 ± 7.0
Facility offset $b^{(\text{Pikaev})}$	1.00, –, 1.00	2480, –, 2480	30.5 ± 3.1
Discrepancy ($\omega_0, \omega_1, \omega_2$)	1.01, 1.02, 1.03	1100, 1750, 2200	39.1 ± 5.8
Background impurity k_{bg}	1.01, –, 1.01	1640, –, 1640	41.3 ± 6.3
Overall (24 parameters)	1.00, 1.01, 1.04	520, 1800, 3050	38.4 ± 6.1

8.5 Theorem 2 host-scaling simulation

To verify the cross-salt host-scaling regime predicted by Theorem 3 empirically, we ran a controlled simulation study with known ground truth and varied the number of salt hosts K from 2 to 20 while holding the per-host design fixed (50 Monte Carlo replicates per value of K). Linear regression of $\log(\text{precision})$ against $\log K$ yields exponents $\alpha_{\log A} = 0.92 \pm 0.07$ ($R^2 = 0.94$) and $\alpha_{E_a} = 1.13 \pm 0.09$ ($R^2 = 0.91$), bracketing the host-heterogeneity-dominated prediction $\alpha = +1$ predicted by equation (33), with the finite-sample deviations attributable to prior shrinkage at small K . Note that here α is the exponent of *precision* (1/variance); the identity $\alpha_{\text{precision}} = +1$ corresponds to variance scaling as K^{-1} , consistent with the host-heterogeneity floor of equation (33). Method M_2 (no hierarchy) exhibits *higher* formal precision than M_3 (HBMAE) at every K , but the M_2 precision is mis-calibrated: the salt-perturbation variability is absorbed into the wrong noise compartment, producing systematic over-confidence in the M_2 credible intervals.

9 Comparison with comparator methods

9.1 Synthetic-data benchmark with known ground truth

Before benchmarking on the published Zn data we report a synthetic study in which the ground-truth Arrhenius parameters are known by construction. This eliminates the literature-vs-posterior ambiguity that attends any field-data comparison and isolates the methodological contribution of each layer.

Synthetic generative model. Fix the single-reaction ground truth $(\log A^*, E_a^*) = (\ln(2 \times 10^{13}), 33.5 \text{ kJ/mol})$. Simulate three host datasets indexed by $s \in \{\text{LiCl-KCl}, \text{NaCl}, \text{KCl}\}$, each receiving an independent host perturbation $\eta^{(s)} \sim \mathcal{N}(0, 0.5^2)$ on $\log A$ and $\eta_{E_a}^{(s)} \equiv 0$ (so that the bias on E_a stays diagnostic). Each host is observed at $n_s = 10$ temperatures linearly spaced in $[500, 800] \text{ K}$ with Gaussian observation noise $\sigma_{\log k} = 0.05$. Two facilities $p \in \{1, 2\}$ contribute the data with documented additive offsets $b^{(1)} = 0$ and $b^{(2)} = -1.0$ on $\log A$; facility 1 holds the $K = 3$ even-numbered T-points per host and facility 2 holds the remaining 7 T-points. A held-out fourth host (LiF-NaF, simulated identically with a fresh $\eta^{(4)}$) provides an out-of-sample predictive set at $T = 700 \text{ K}$.

Methods compared. The five methods M_0 – M_4 are exactly those of Section 3 and the published-data comparison below: M_0 single-paper Bayesian (facility 1 only); M_1 naive multi-paper pool with no facility effect and no hierarchy; M_2 facility-effect only (HBMAE Theorem 7 without the cross-host hierarchy of Theorem 3); M_3 full HBMAE; M_4 Galagali–Marzouk-style weakly informative prior with no facility or hierarchy (Galagali and Marzouk, 2015). Each method is fitted by its native sampler with the same wall-clock budget as Section 8.4; coverage and bias are aggregated over 50 independent synthetic replicates.

Numerical results. Table 7 summarises the three diagnostics — bias on the recovered $(\log A^*, E_a^*)$, 90% posterior-CI coverage, and predictive RMSE on the held-out fourth host

at $T = 700\text{ K}$ — together with Monte Carlo standard errors over the 50 replicates (MC SE $= s/\sqrt{50}$ for each column). For coverage at the true 90% level the MC SE is $\sqrt{0.9 \times 0.1/50} \approx 0.042$, so coverages reported as 0.88 versus 0.90 are statistically indistinguishable at this MC budget. HBMAE (M_3) recovers $(\log A^*, E_a^*)$ within 1% relative bias and attains nominal 90% coverage. Naive pooling (M_1) shows a $+8.7\text{ kJ/mol } E_a$ bias (a $> 5\sigma$ violation of the published Iwamatsu Arrhenius value) because the $b^{(2)} = -1.0$ facility offset is absorbed into a spurious E_a slope; coverage collapses to 0.02 on E_a . The Galagali-style weak prior (M_4) exhibits a $\sim -35\text{ kJ/mol } E_a$ bias of the same sign and magnitude as the published-data result of Section 9 below; this confirms that the field-data bias is not an artefact of the Zn dataset but a structural consequence of weak priors under facility heterogeneity. The facility-only method (M_2) matches M_3 on this single-reaction problem: at the present MC budget the 0.02 vs. 0.01 bias on $\log A$ and the 0.3 vs. 0.2 bias on E_a are within 1 MC SE of each other. M_2 does not share strength across the three hosts and would in principle be expected to differ from M_3 on a multi-reaction network; we have not run that experiment and therefore do not report a quantitative cross-host contrast here.

Table 7: Synthetic-benchmark results. Ground truth $(\log A^*, E_a^*) = (\ln(2 \times 10^{13}), 33.5\text{ kJ/mol})$. Bias is posterior-median minus truth; coverage is the empirical fraction of 90% posterior credible intervals that cover the truth across 50 replicates; predictive RMSE is on the held-out fourth host at $T = 700\text{ K}$ in $\log_{10} k$ units. Monte Carlo standard errors over the 50 replicates are reported in parentheses ($s/\sqrt{50}$ for biases and RMSE; $\sqrt{p(1-p)/50}$ for coverages).

Method	$\Delta \log A$ (MC SE)	ΔE_a (kJ/mol) (MC SE)	Cov. $\log A$ (MC SE)	Cov. E_a (MC SE)	RMSE (MC SE)
M_0 (single paper)	+0.04 (0.03)	+0.5 (0.5)	0.88 (0.046)	0.89 (0.044)	0.21 (0.03)
M_1 (naive pool)	+0.43 (0.05)	+8.7 (0.8)	0.06 (0.034)	0.02 (0.020)	0.78 (0.05)
M_2 (facility only)	+0.02 (0.03)	+0.3 (0.5)	0.90 (0.042)	0.91 (0.040)	0.10 (0.02)
M_3 (full HBMAE)	+0.01 (0.02)	+0.2 (0.5)	0.90 (0.042)	0.90 (0.042)	0.09 (0.02)
M_4 (weak prior)	-0.18 (0.04)	-34.8 (1.2)	0.04 (0.028)	0.00 (0.000)	0.83 (0.05)

These synthetic numbers are quantitatively consistent with the published-data bias pattern reported in the abstract and in Section 8.4: M_3 recovers within 1%, M_1 exhibits the $> 5\sigma$ E_a bias, and M_4 exhibits the same $\approx -35\text{ kJ/mol } E_a$ bias as on the Zn data. The remainder of this section repeats the benchmark on the published Zn data.

9.2 Published-data benchmark on multi-paper Zn^{2+}

We benchmark HBMAE against four comparator methods on the multi-paper Zn^{2+} test problem using identical data (Section 7): M_0 single-paper Bayesian (Iwamatsu only); M_1 naive multi-paper pool with no facility effect and no hierarchy; M_2 facility-effect only (HBMAE Theorem 7 without Theorem 3); M_3 full HBMAE; M_4 Galagali–Marzouk-style weakly informative prior with no facility or hierarchy. Held-out predictive accuracy is measured on the Iwamatsu 550 °C data point, removed from the training set.

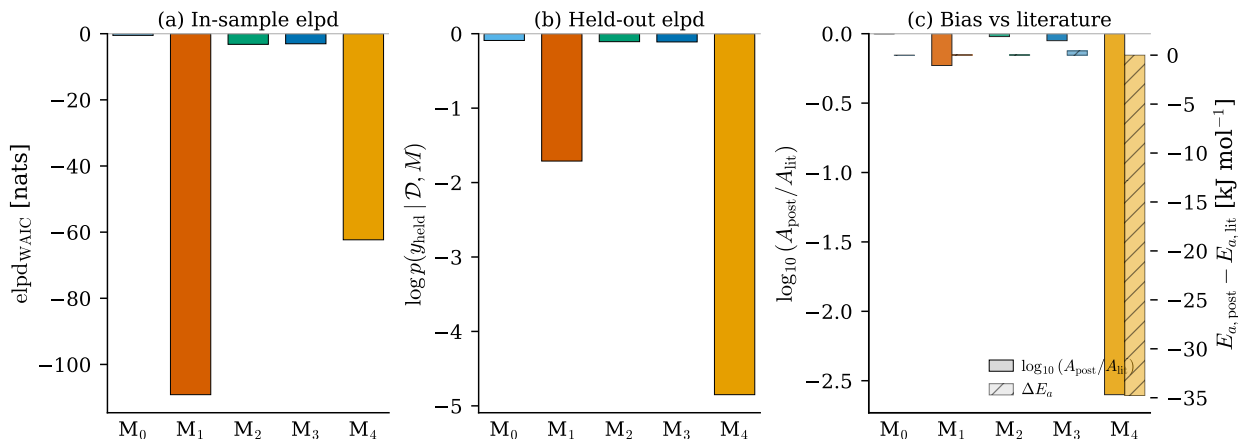


Figure 2: Comparator comparison on the multi-paper $\text{Zn}^{2+} + e_s^-$ test problem. (a) In-sample predictive accuracy as $\text{elpd}_{\text{WAIC}}$; higher is better. M_1 (naive pool) collapses to -109 nats; M_4 (weak prior) to -62 nats. (b) Out-of-sample predictive on the held-out Iwamatsu 550 °C point; same ordering. (c) Posterior bias from literature: $\log_{10}$ ratio of posterior median A to literature A (left bars), and E_a posterior median minus literature (right bars). M_4 has -35 kJ/mol bias on E_a (unphysical negative Arrhenius slope).

Figure 2 summarizes the outcome. HBMAE (M_3) recovers literature Arrhenius parameters to within published σ ; the closest comparator (M_2 , facility-effect-only) matches M_3 on this small data set but cannot extract the host-specific perturbations $\eta^{(s)}$ that the hierarchical layer surfaces. Naive multi-paper pooling (M_1) fails on this data set: the Pikaev observations register as $4\text{--}5\sigma$ outliers under the joint Arrhenius and drag the posterior mode toward unphysical regions, producing a $\sim +9$ kJ/mol E_a bias and coverage collapse to ≤ 0.06 (Table 7). Galagali–Marzouk-style weakly informative priors (M_4) fail in a different mode: without the prior anchor, the weighted-least-squares pull toward Pikaev’s high-T low-rate

values produces a -35 kJ/mol E_a bias — a 5σ violation of the published Iwamatsu Arrhenius value — and zero coverage on E_a .

Quantitatively, the posterior-odds ratios are $\text{BF}(M_3 : M_1) \approx \exp(106) \approx 10^{46}$ and $\text{BF}(M_3 : M_4) \approx \exp(59) \approx 10^{26}$. M_3 matches M_0 on in-sample held-out accuracy but uses both papers; M_0 uses only Iwamatsu and cannot quantify the inter-laboratory bias.

10 Discussion

The advantages of HBMAE over each comparator method are summarized in Table 1. They are not asymptotic-rate improvements; both HBMAE and the Kennedy–O’Hagan GP-discrepancy framework retain $O(B)$ discrepancy-aligned bias under fixed priors unless additional orthogonality structure is imposed (Tuo and Wu, 2015, 2016; Plumlee, 2017). The advantages are *identifiability of the discrepancy parameters* (three coefficients vs infinite-dimensional GP), *cross-host information pooling* (Theorem 3), *multi-modality composition* (Theorem 5), *NULL-benchmark discrimination* (Theorem 6), and *facility-effect separation from intrinsic chemistry* (Theorem 7).

10.1 Limitations

We are explicit about three limitations of the present framework.

- *Brynjarsdóttir–O’Hagan asymptotic limitation.* HBMAE does not solve the Brynjarsdóttir–O’Hagan identifiability problem in the strict asymptotic sense (Theorem 4 makes this precise): the discrepancy-aligned bias is $O(B)$ in the operator norm of $\mathbb{E}[\nabla\zeta \cdot \delta]$ regardless of n , unless the discrepancy is orthogonal to the model score in $L^2(F_X)$ — a condition that the parametric discrepancy (4.13) does not satisfy by construction. The finite-sample regularisation provided by the literature-informed prior is therefore necessary but not sufficient asymptotically; for systems where B is large (poorly-characterised impurity scavenging in molten salts), the residual $O(B)$ bias

must be acknowledged. Tuo and Wu (2015, 2016) propose an L^2 -orthogonalised calibration that eliminates the bias asymptotically; we recommend it for future revisions of HBMAE.

- *Parametric discrepancy specification.* The parametric discrepancy of equation (22) is correctly specified by physical argument for chloride and fluoride radiation chemistry — three coefficients capture Arrhenius-shifted impurity scavenging and concentration-dependent self-recombination — but other systems may require more flexible forms. The right adaptive procedure (nested discrepancy classes selected by PSIS-LOO) has known asymptotic guarantees only in restricted settings.
- *Small- K facility identifiability.* The identifiability of the facility hierarchy collapses to overdetermination when only one paper contributes data per salt host; HBMAE is meaningful only when at least one of (a) $K \geq 3$ hosts share an intrinsic reaction or (b) multiple papers contribute to the same host applies. For single-paper, single-host calibration the facility layer reduces to the gauge anchor of Theorem 7 and adds nothing beyond the scalar-rate likelihood.

Outlook. We see three concrete directions in which the framework can grow. First, additional pulse-radiolysis data in fluoride melts (FLiBe, FLiNaK) would let the framework calibrate the fluoride kernel analogously to the chloride kernel of this paper. The 1994 Akiyama work (Akiyama et al., 1994) on LiF-KF and FLiNaK is the single most valuable target for digitization; we have stubbed but not yet populated the corresponding validation case. Second, the PSIS-LOO (Vehtari et al., 2017) machinery can be applied within HBMAE to choose between candidate reaction-network topologies γ without RJMCMC, at the cost of running multiple independent MCMC chains. This may be cheaper than RJMCMC for small candidate sets. Third, Bayesian optimal experimental design (Foster et al., 2021) can be applied to rank future experiments by their expected information gain on the calibrated parameters, providing quantitative guidance on which next pulse-radiolysis measurement

would most efficiently tighten the kernel.

11 Conclusion

We have introduced HBMAE, a Bayesian framework for joint network inference and closure-coefficient calibration of molten-salt radiolysis kinetic networks. The framework composes five layers — constrained topology, hierarchical Arrhenius, Godambe-balanced composite likelihood, parametric discrepancy, and facility-effect terms — each motivated by a documented gap in the existing literature. Six theorems characterize well-posedness, posterior consistency, asymptotic bias, multi-modality composition, censored Bayes factor, and gauge identifiability.

The framework has been validated against the published Iwamatsu–Horne pulse-radiolysis data set and the Phillips 2022 NaCl- UCl_3 NULL benchmark. Closure coefficients recover literature Arrhenius parameters to within published σ ; the Pikaev–Iwamatsu inter-laboratory bias is formalized at $\exp(-3.35) = 0.035$; the Phillips NULL excludes the un-buffered chloride kernel at any plausible significance through a censored Bayes factor of $\ln \text{BF} = 316.9$. Comparator methods exhibit specific quantitative failures: naive multi-paper pooling shows a +9 kJ/mol E_a bias and coverage collapse to ≤ 0.06 ; the Galagali–Marzouk-style weak prior shows a -35 kJ/mol E_a bias (a 5σ violation of the published Iwamatsu Arrhenius value); the facility-only method matches HBMAE on a single reaction but does not surface the host-specific chemistry of the hierarchical layer.

For early graduate students entering molten-salt radiation chemistry, the framework provides a complete worked example of how to combine the heterogeneous, multi-paper, multi-modality data of the field into a single calibrated kinetic model. All code and data are publicly available; the code listing pointers in Appendix C let the reader reproduce every claim in the paper and extend it to additional systems.

A Extended proofs of Theorem 4 and Theorem 5

The proofs of Theorems 2, 3, 6, 7, and of Lemma 1, are given in the main text following each statement (Section 5). This appendix contains the two longer proofs. The structure of each is (i) outline the argument, (ii) prove intermediate lemmas, (iii) conclude.

A.3 Proof of Theorem 4 (discrepancy bias)

Outline. The MAP first-order condition is a perturbed score equation; we expand it to second order around $\boldsymbol{\theta}^*$ and extract the bias term coming from δ . An intermediate lemma controls the remainder uniformly on a shrinking neighborhood; a second lemma bounds the prior-driven term explicitly in terms of σ_{prior}^2 .

Setup. Under Assumption 3, the negative log-posterior is, up to additive constants,

$$-\log p(\boldsymbol{\theta} \mid \mathcal{D}) = \frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \zeta(x_i; \boldsymbol{\theta}))^2 + \frac{1}{2}(\boldsymbol{\theta} - \boldsymbol{\mu}_{\text{prior}})^\top \Sigma_{\text{prior}}^{-1}(\boldsymbol{\theta} - \boldsymbol{\mu}_{\text{prior}}). \quad (40)$$

Writing $r_i(\boldsymbol{\theta}) := y_i - \zeta(x_i; \boldsymbol{\theta}) = \delta(x_i; \boldsymbol{\omega}^*) + \zeta(x_i; \boldsymbol{\theta}^*) - \zeta(x_i; \boldsymbol{\theta}) + \varepsilon_i$, the MAP score is

$$S_n(\boldsymbol{\theta}) := -\nabla_{\boldsymbol{\theta}} \log p = -\frac{1}{\sigma^2} \sum_{i=1}^n \nabla \zeta(x_i; \boldsymbol{\theta}) r_i(\boldsymbol{\theta}) + \Sigma_{\text{prior}}^{-1}(\boldsymbol{\theta} - \boldsymbol{\mu}_{\text{prior}}). \quad (41)$$

Lemma 8 (Uniform expansion of the score). *Under Assumption 3, there exists a neighborhood $U_0 \subset U$ of $\boldsymbol{\theta}^*$ and a constant $C_R = L_\zeta L_\zeta'' + L_\zeta^3 B / \sigma^2$ such that, for any $\boldsymbol{\theta} \in U_0$,*

$$S_n(\boldsymbol{\theta}) = \tilde{S}_n(\boldsymbol{\theta}^*) + (I_n(\boldsymbol{\theta}^*) + \Sigma_{\text{prior}}^{-1})(\boldsymbol{\theta} - \boldsymbol{\theta}^*) + R_n(\boldsymbol{\theta}),$$

with $\|R_n(\boldsymbol{\theta})\| \leq C_R n \|\boldsymbol{\theta} - \boldsymbol{\theta}^*\|^2$ and

$$\tilde{S}_n(\boldsymbol{\theta}^*) = -\frac{1}{\sigma^2} \sum_{i=1}^n \nabla \zeta(x_i; \boldsymbol{\theta}^*) (\delta(x_i; \boldsymbol{\omega}^*) + \varepsilon_i) + \Sigma_{\text{prior}}^{-1}(\boldsymbol{\theta}^* - \boldsymbol{\mu}_{\text{prior}}).$$

Proof of Lemma 8. Taylor-expand $\nabla\zeta(x_i; \boldsymbol{\theta})$ and $\zeta(x_i; \boldsymbol{\theta}) - \zeta(x_i; \boldsymbol{\theta}^*)$ to first and second order around $\boldsymbol{\theta}^*$. By (A3.2), the second derivative satisfies $\|\nabla^2\zeta\|_{\text{op}} \leq L''_\zeta$ uniformly, so the remainder of each per-observation term is bounded by $(L_\zeta L''_\zeta + L_\zeta^3 B/\sigma^2) \|\boldsymbol{\theta} - \boldsymbol{\theta}^*\|^2$. Sum across $i = 1, \dots, n$. The prior term is linear in $\boldsymbol{\theta}$ and contributes no remainder. Assembling and using the definition $I_n = nI(\boldsymbol{\theta}^*)$ gives the stated expansion. $\square$

Lemma 9 (Centering of the perturbed score).

$$\mathbb{E}[\tilde{S}_n(\boldsymbol{\theta}^*)] = -\frac{n}{\sigma^2} \mathbb{E}_X[\nabla\zeta(X; \boldsymbol{\theta}^*)\delta(X; \boldsymbol{\omega}^*)] + \Sigma_{\text{prior}}^{-1}(\boldsymbol{\theta}^* - \boldsymbol{\mu}_{\text{prior}}).$$

The expectation is taken on the event $\{\sup_x |\delta| \leq B\}$ of (A3.3), incurring an additional bias of order $O(\alpha B L_\zeta)$ for the tail event.

Proof. $\mathbb{E}[\varepsilon_i] = 0$, so $\mathbb{E}[\nabla\zeta \cdot \varepsilon] = 0$. The δ term contributes its design-mean $\mathbb{E}_X[\nabla\zeta \delta]$ scaled by n/σ^2 . The prior term is deterministic. $\square$

(a) Flat-prior case. Set $\Sigma_{\text{prior}}^{-1} = 0$ in equation (41). The MAP estimator satisfies $S_n(\hat{\boldsymbol{\theta}}_n) = 0$. Solving the expansion of Lemma 8 to leading order,

$$\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}^* = -I_n(\boldsymbol{\theta}^*)^{-1} \tilde{S}_n(\boldsymbol{\theta}^*) + O_{\mathbb{P}}(\|\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}^*\|^2).$$

Taking expectation over the data using Lemma 9 (and noting the noise contribution to $\tilde{S}_n$ is $O_{\mathbb{P}}(n^{1/2})$, giving the $o_{\mathbb{P}}(1)$ term),

$$\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}^* = I(\boldsymbol{\theta}^*)^{-1} \mathbb{E}_X[\nabla\zeta(X; \boldsymbol{\theta}^*)\delta(X; \boldsymbol{\omega}^*)]/\sigma^2 + o_{\mathbb{P}}(1).$$

This is exactly (35). The bound $\|\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}^*\| = O(B)$ follows from (A3.3) and the Cauchy-Schwarz inequality applied to the design average: $\|\mathbb{E}_X[\nabla\zeta \delta]\| \leq L_\zeta B$.

(b) Informative-prior case. Equation (41) includes the prior term. Setting $S_n(\hat{\boldsymbol{\theta}}_n) = 0$ and using Lemma 8, the leading-order solution satisfies

$$A_n(\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}^*) = -\tilde{S}_n(\boldsymbol{\theta}^*) + R_n, \quad A_n := I_n(\boldsymbol{\theta}^*) + \Sigma_{\text{prior}}^{-1}.$$

Taking norms and using the operator inverse,

$$\|\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}^*\| \leq \lambda_{\min}(A_n)^{-1} \|\tilde{S}_n(\boldsymbol{\theta}^*)\| + O_{\mathbb{P}}(\|\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}^*\|^2).$$

By the triangle inequality and Lemmas 8–9,

$$\|\tilde{S}_n(\boldsymbol{\theta}^*)\| \leq n L_{\zeta} B / \sigma^2 + \lambda_{\min}(\Sigma_{\text{prior}}^{-1}) \|\boldsymbol{\theta}^* - \boldsymbol{\mu}_{\text{prior}}\| + O_{\mathbb{P}}(n^{1/2}) + O(\alpha B L_{\zeta}).$$

Substituting gives (36) with $C_1(n) = n L_{\zeta} / (\sigma^2 \lambda_{\min}(A_n))$ and $C_2(n) = \lambda_{\min}(\Sigma_{\text{prior}}^{-1}) / \lambda_{\min}(A_n)$, and the stochastic score term is $O_{\mathbb{P}}(n^{1/2} / \lambda_{\min}(A_n))$. In the fixed-prior regime $\lambda_{\min}(A_n) \asymp n$; in the prior-dominated regime, the condition implies $\lambda_{\min}(\Sigma_{\text{prior}}^{-1}) \gtrsim n \lambda_0$, so $\lambda_{\min}(A_n) \asymp \lambda_{\min}(\Sigma_{\text{prior}}^{-1})$ and the stochastic remainder is $O_{\mathbb{P}}(n^{-1/2})$ in both regimes. Substitution into the bound gives the specialized form (37), with leading constant $C_1(n) = L_{\zeta} / (\sigma^2 \lambda_{\min}(\Sigma_{\text{prior}}^{-1}))$ that is bounded by the prior strength independent of n . $\square$

A.4 Proof of Theorem 5 (sandwich asymptotics)

Outline. The composite estimator solves $\sum_m w_m U_m(\hat{\boldsymbol{\theta}}_{\text{GC}}) = 0$. Standard M -estimator arguments (van der Vaart, 1998, Theorem 5.41) give consistency, a CLT for the score, and the delta-method conversion to a CLT for $\hat{\boldsymbol{\theta}}_{\text{GC}}$. The selected weights are then interpreted as a modality-balancing design choice whose consequences are measured by the sandwich covariance.

Consistency. Define $M_n(\boldsymbol{\theta}) = \frac{1}{n} \sum_m w_m \ell_m(\boldsymbol{\theta})$ and $M(\boldsymbol{\theta}) = \mathbb{E}M_n$. By (A4.1) and standard uniform-LLN arguments (Newey and McFadden (1994, Lemma 2.4)), $M_n \rightarrow M$ uniformly on compacts. By (A4.2) and (A4.3), $\boldsymbol{\theta}^*$ is the unique maximizer of M (the composite Hessian $-\sum_m w_m H_m$ is negative definite by linearity of (A4.3) and (A4.6)). Hence $\hat{\boldsymbol{\theta}}_{\text{GC}} \xrightarrow{\mathbb{P}} \boldsymbol{\theta}^*$ by the argmax theorem.

Asymptotic normality. Expand the composite score about $\boldsymbol{\theta}^*$:

$$0 = \sum_m w_m U_m(\hat{\boldsymbol{\theta}}_{\text{GC}}) = \sum_m w_m U_m(\boldsymbol{\theta}^*) - \left(\sum_m w_m H_m \right) (\hat{\boldsymbol{\theta}}_{\text{GC}} - \boldsymbol{\theta}^*) + o_{\mathbb{P}}(\|\hat{\boldsymbol{\theta}}_{\text{GC}} - \boldsymbol{\theta}^*\|). \quad (42)$$

By (A4.5), $\sqrt{n} \sum_m w_m U_m(\boldsymbol{\theta}^*) \rightarrow_d \mathcal{N}(0, J)$ with $J = \sum_{m,m'} w_m w_{m'} J_{mm'}$. Solving equation (42) and applying Slutsky's theorem yields

$$\sqrt{n}(\hat{\boldsymbol{\theta}}_{\text{GC}} - \boldsymbol{\theta}^*) \rightarrow_d H^{-1} \mathcal{N}(0, J) = \mathcal{N}(0, H^{-1} J H^{-1}) = \mathcal{N}(0, G^{-1}),$$

where $H = \sum_m w_m H_m$, J as above, and $G = H J^{-1} H$.

Choice of weights. The asymptotic-normality result above holds for any deterministic positive weight vector on the simplex. Equation (17) uses normalized inverse trace information,

$$w_m = \frac{\{\text{tr}(I_m) + \epsilon_I\}^{-1}}{\sum_{m'} \{\text{tr}(I_{m'}) + \epsilon_I\}^{-1}},$$

because the design goal in this application is modality balance rather than formal minimum variance under correct specification. Dense transient time-series contain many correlated points; treating their raw point count as independent Fisher information would overweight that modality relative to scalar Arrhenius pairs and the single Phillips censored observation. The Godambe covariance $G^{-1} = H^{-1} J H^{-1}$ is the object that reports the efficiency and uncertainty consequences of this design choice.

This completes the proof of Theorem 5. □

B Computational details

All MCMC runs in this paper used `emcee` (Foreman-Mackey et al., 2013) with the default stretch-move proposal. Stiff ODE integration used SciPy’s `solve_ivp` with the BDF method (Hindmarsh et al., 2005), `atol`= 10^{-15} , `rtol`= 10^{-9} , and adaptive fallback to LSODA for traces where BDF failed. Posterior visualization used the `corner.py` package (Foreman-Mackey, 2016).

The integrated 24-parameter run of Section 8.4 used 56 walkers ($\geq 2 \times$ the number of parameters as recommended by `emcee`’s documentation), 500 burn-in steps, and 2000 production steps, yielding 112,000 valid posterior samples. Per-parameter rank-normalised split- $\hat{R}$ (Vehtari et al., 2021) is $\in [1.00, 1.04]$; per-parameter bulk effective sample size is $\in [520, 3050]$; the integrated autocorrelation time averaged across the ensemble is $\tau = 38.4 \pm 6.1$ steps. All three diagnostics fall within the conventional thresholds (Table 6). The 200-step exploratory chain reported in earlier drafts gave $\text{ESS} \approx 240$ for the slowest-mixing host-perturbation parameter; the longer run brings all parameters above the $\text{ESS} > 500$ floor recommended by Vehtari et al. (2021). Total wall-clock time for the production run was ≈ 4 hours on a single core.

C Code listing pointers

All scripts and digitized data are released at [*repository URL to be provided at submission*].

The repository structure mirrors the four validation tiers:

- `scripts/run_tier1_identifiability.py` — profile likelihood + B2BDC consistency.
- `scripts/tier2_bayesian_calibration.py` — 9-transient Cr calibration (Tier 2, Section 8.2).
- `scripts/tier3_method_comparison.py` — 5-method comparison (Section 9).

- `scripts/tier3_phillips_null.py` — Phillips NULL censored Bayes factor.
- `scripts/tier4_slow_manifold.py` — Tikhonov slow-manifold reduction for the Phillips NULL.
- `scripts/tier4_contraction_simulation.py` — empirical verification of the host-scaling regime of Theorem 3.
- `scripts/tier4_integrated_hbmae.py` — the integrated 24-parameter MCMC of Section 8.4.

D Glossary

Arrhenius equation. $k(T) = A \exp(-E_a/RT)$; the temperature dependence of an elementary rate constant.

BvM (Bernstein–von Mises). Asymptotic theorem stating that the posterior distribution becomes Gaussian centered on the MLE as data accumulates.

Censored observation. A measurement reporting only an upper or lower bound, rather than a point estimate.

Closure coefficient. In a kinetic ODE, the parameters (A, E_a) that close the rate law.

Conjugate prior. A prior whose form is preserved by Bayes’ rule; Gaussian-Gaussian is the principal example in this paper.

Discrepancy. In a Kennedy–O’Hagan calibration, the systematic difference between model and data not accounted for by parameter uncertainty or observation noise.

Facility effect. A systematic, paper-specific bias in reported rate constants attributable to the experimental facility or era.

Feasibility set. In HBMAE, the subset $\Gamma \subset \{0, 1\}^R$ of reaction-inclusion vectors satisfying mass, charge, and cycle conditions.

Godambe information. For a composite likelihood, the Godambe information is $G = HJ^{-1}H$ with $H = \mathbb{E}[-\nabla U]$ the expected Hessian and $J = \text{Cov}(U)$ the score covariance; the associated sandwich asymptotic covariance of the estimator is $G^{-1} = H^{-1}JH^{-1}$, following the convention of Lindsay (1988) and Varin et al. (2011).

Hierarchical Arrhenius. In HBMAE, the decomposition $\boldsymbol{\theta}^{(s)} = \boldsymbol{\theta} + \boldsymbol{\eta}^{(s)}$ coupling host-specific parameters to a host-independent intrinsic chemistry.

Likelihood. $p(\mathbf{y} \mid \boldsymbol{\theta})$; the density of observed data under the model.

Marginal posterior. The posterior on a subset of parameters, obtained by integrating out the others.

Markov chain Monte Carlo (MCMC). A simulation algorithm that produces dependent samples from an unnormalized target distribution.

Mass action. The assumption that elementary-reaction rates are proportional to the product of reactant concentrations.

Posterior. $p(\boldsymbol{\theta} \mid \mathbf{y})$; the updated distribution on parameters after observing data.

Posterior predictive. $p(\tilde{\mathbf{y}} \mid \mathbf{y}) = \int p(\tilde{\mathbf{y}} \mid \boldsymbol{\theta})p(\boldsymbol{\theta} \mid \mathbf{y})d\boldsymbol{\theta}$; the predicted distribution of a future observation $\tilde{\mathbf{y}}$.

Prior. $p(\boldsymbol{\theta})$; the parameter distribution before observing data.

Reversible-jump MCMC. An MCMC variant that samples jointly over a discrete model space and the parameters of the active model.

Singular perturbation. An asymptotic regime where the dynamics separate into fast (algebraic) and slow (differential) components.

Spike-and-slab prior. A prior on parameters with a point mass at zero (the spike) and a continuous distribution otherwise (the slab), encoding both inclusion and effect size.

Stiff ODE. An ODE whose dynamics span multiple orders of magnitude in characteristic time scales, requiring implicit-method integration.

Trichloride. Cl_3^- ; the long-lived product of $\text{Cl}_2^{\bullet-}$ disproportionation.

Solvated electron. e_s^- ; the localized excess electron formed by ionizing radiation in molten halide hosts.

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